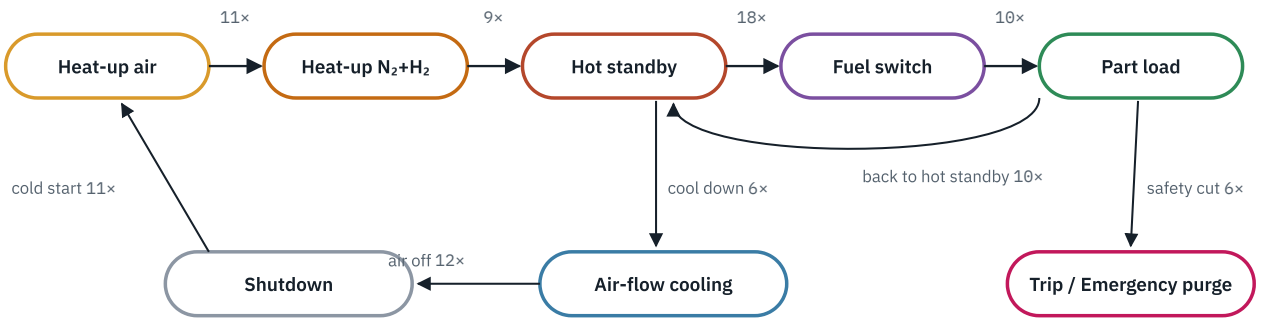
Stage atlas

One page for each of the 8 plant stages, and two pages that show **what changes, how much and how fast** when the plant moves from one stage to the next. Every number is a median of the real plant runs in the 39 logged days. Each stage page starts with the **plant layout**, with the groups that work in that stage lit up and numbered.



The plant's main path. Numbers are how many times each move happened between two stretches of logged data. Click a stage to open its page.

The 8 stages

STAGE 1

Heat-up air

Start-up with hot air only. 14 visits, 56.3 h.

STAGE 2

Heat-up N_2+H_2

Start-up with forming gas. 15 visits, 81.5 h.

STAGE 3

Hot standby

Hot and ready, no power. 28 visits, 338 h.

STAGE 4

Fuel switch

Natural gas on, no current yet. 19 visits, 3.3 h.

STAGE 5

Part load

Making power. 10 visits, 4.5 h.

STAGE 6

Air-flow cooling

Shutting down, air still blowing. 28 visits, 254.9 h.

STAGE 7

Shutdown

Everything off except the controls. 33 visits, 173.7 h.

STAGE 8

Trip / Emergency purge

Safety stop. 6 visits, 1.3 h.

Every move between stages

Rows are the stage the plant left, columns the stage it entered (numbered as the cards above). Click a number to see that change. Grey numbers happened only once and are not drawn.

FROM ↓ · TO →	1	2	3	4	5	6	7	8
1 Heat-up air	·	11	·	·	·	1	2	·
2 Heat-up N_2+H_2	1	·	9	·	·	1	3	·
3 Hot standby	·	·	·	18	·	6	1	1
4 Fuel switch	·	·	7	·	10	·	1	1
5 Part load	·	·	3	1	·	3	1	2
6 Air-flow cooling	8	1	·	·	·	·	12	2
7 Shutdown	3	2	7	·	·	14	·	·
8 Trip / Emergency purge	·	·	1	·	·	2	3	·

The two change pages

PAGE 10

Starting up and loading

8 moves: cold start, forming gas on, holding hot, natural gas on, current on.

PAGE 11

Unloading, cooling, trips

12 moves: gas off, cool-down, air off, aborted starts, safety cuts.

If you remember one thing: Each stage has its own page; the change pages show which readings jump at once and which take hours to follow.

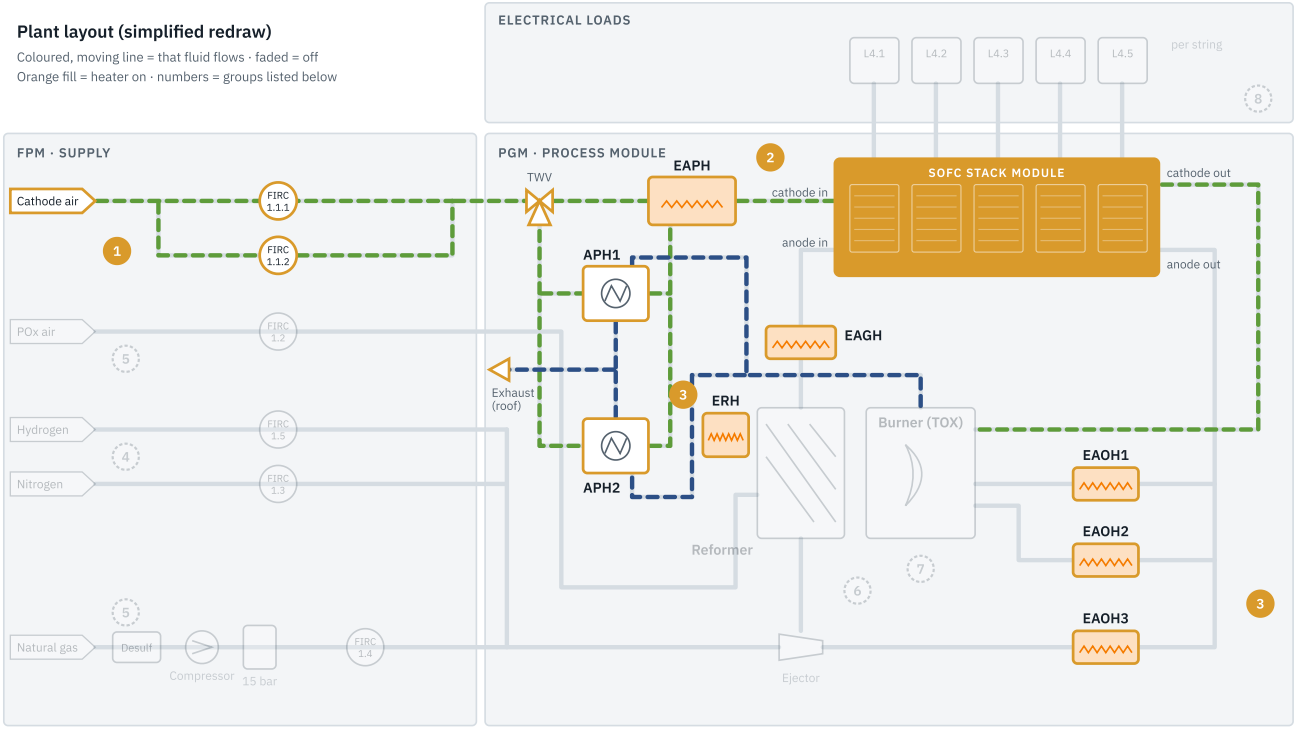
STAGE 1 OF 8 · START-UP WITH HOT AIR ONLY

1 Heat-up air

> The stack is cold. Two air streams blow 6000 ln/min through it and the electric air heater warms that air. Nothing flows on the fuel side yet.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
14	56.3 h	3.9 h	9.6 h

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000

2 Electric air heater EAPH
On · 13 kW

3 Gas-line heaters ERH · EAGH · EAOH 1–3
On · 3 A

4 Forming gas N_2 + H_2
Off

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

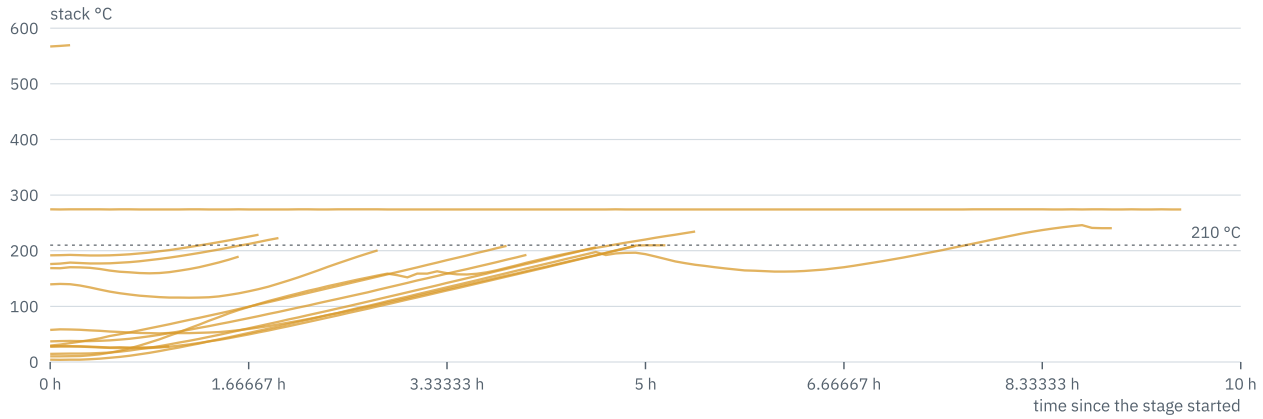
Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000
Electric air heater EAPH · 13 kW
Recuperators APH (passive)
Gas-line heaters ERH, EAGH, EAOH 1–2 · 3 A
Recycle heater EAOH 3 · 8 A
~~Forming gas N_2 1200 + H_2 60~~
~~Natural gas + reformer air~~
~~Reformer + gas recycle hot~~
~~After-burner burning~~
~~Stack current to loads~~
Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

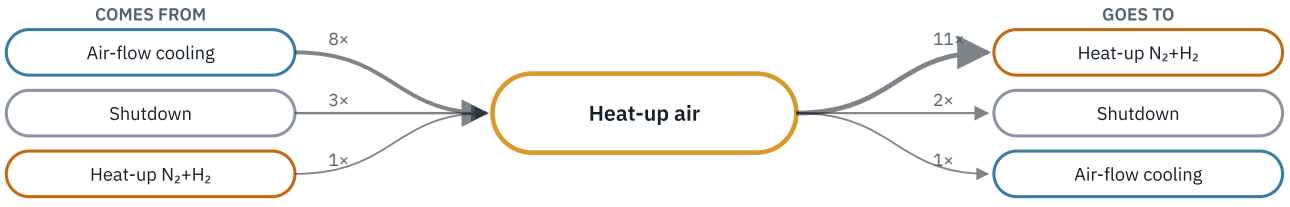


Every visit of this stage drawn from its start (one line per visit, 14 visits; shown up to 10.0 h). Stack warms steadily, about +42 °C/h (limit 120 °C/h). The heaters follow the air: they come on 0–9 min after the air starts.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**159** °C	33.7 – 274
Air into stack (TIRC 2.06)	**204** °C	57.2 – 280
Cathode air, both streams	**6,000** ln/min	5,999 – 9,998
Nitrogen N_2 (FIRC 1.3)	**0.0** ln/min	0.0 – 0.0
Hydrogen H_2 (FIRC 1.5)	**0.0** ln/min	0.0 – 0.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**1.5** V	0.60 – 9.9
Air heater EAPH power	**12.6** kW	1.1 – 16.7
Gas-line heater ERH current	**3.2** A	0.0 – 4.7
Recycle heater EAOH 3 current	**8.2** A	0.0 – 10.9
After-burner (TIRC 5.09)	**142** °C	30.8 – 271
Reformer catalyst inlet (TIRC 3.07)	**124** °C	24.7 – 267
Exhaust safety (TIRZ 8.1.10)	**28.3** °C	20.0 – 61.9

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. Typical visit ≈ 4 h (longest 9.5 h). Next: Heat-up N_2+H_2 in 11 of 14 cases.

Shutdown → Heat-up air | Air-flow cooling → Heat-up air | Heat-up air → Heat-up N_2+H_2 | Heat-up air → Shutdown

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,000** ln/min	2,999 – 4,999
FIRC 1.1.2	Cathode air stream 2	**3,000** ln/min	3,000 – 4,999
PIR 1.1.1	Air stream 1 supply	**5.0** bar	4.8 – 5.1
PIR 1.1.3	Air stream 2 supply	**4.9** bar	4.7 – 5.0
TIR 1.1.1	Air after MFCs	**19.2** °C	15.2 – 20.1
PIR 1.1.2	Air after MFCs	**85.8** mbar	74.3 – 238
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**154** m^3/h	48.7 – 257
TIR 2.04	TWV → APH inlet	**18.6** °C	15.1 – 19.5
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.4** bar	5.3 – 5.9
TIR 1.2.1	POx air (FPM)	**16.6** °C	9.0 – 17.8
TIR 1.2.2	POx air FPM → PGM	**7.3** °C	-0.01 – 10.2
PIR 1.2.2	POx air after MFC	**2.4** mbar	1.8 – 9.9
TIR 3.04	POx air PGM inlet	**8.4** °C	2.1 – 15.4
TIR 3.05	Reformer inlet POx air	**75.7** °C	20.9 – 180
Nitrogen			
FIRC 1.3	Nitrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.3.1	N_2 supply	**5.6** bar	5.5 – 5.8
TIR 1.3	Purge gas	**15.6** °C	7.3 – 17.0
PIR 1.3.2	Purge gas	**0.48** bar?	0.01 – 0.72
Hydrogen			
FIRC 1.5	Hydrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.5	H_2 supply	**5.2** bar	5.1 – 5.5

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**0.03** bar	0.02 – 12.9
TIR 1.4.1	Fuel (FPM)	**16.8** °C	8.2 – 18.0
PIR 1.4.2	Fuel after MFC	**0.00** bar	0.00 – 0.01
TIR 1.4.2	Fuel FPM → PGM	**7.7** °C	0.27 – 10.8
TIR 3.01	Fuel PGM inlet	**9.0** °C	2.4 – 15.9
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**17.9** °C	14.8 – 18.9
TIR 2.02	APH1+2 outlet air	**119** °C	28.2 – 258
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**140** °C	33.5 – 270
PIR 7.1.1	APH inlet exhaust 1	**1.9** mbar	1.6 – 5.2
TIR 7.3.1	Flue gas out 1	**23.4** °C	17.8 – 48.4
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**17.7** °C	14.5 – 18.6
TIR 2.03	APH outlet riser → EAPH	**118** °C	27.6 – 258
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**140** °C	33.2 – 269
PIR 7.1.2	APH inlet exhaust 2	**1.8** mbar	1.5 – 5.1
TIR 7.3.2	Flue gas out 2	**23.3** °C	17.6 – 47.4
Air heater EAPH			
TIR 2.05	EAPH inlet	**119** °C	28.3 – 257
PIR 2.01	EAPH inlet	**29.8** mbar	19.5 – 81.2
EAPH power	Heater power (3-phase)	**12.6** kW	1.1 – 16.7
TIRZ 8.1.1	EAPH safety (trip 1050)	**189** °C	49.5 – 254
TIRC 2.06	EAPH outlet = stack air in	**204** °C	57.2 – 280
PIR 2.02	Stack module air inlet	**4.5** mbar	3.4 – 14.1

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**16.3** °C	4.0 – 27.1
PIR 3.01	Ejector main stream	**0.00** bar	0.00 – 0.01
TIR 6.01	Recycle gas inlet	**154** °C	59.4 – 218
PIR 6.01	Recycle gas inlet	**2.6** mbar	2.1 – 9.8
TIRC 3.03	Ejector outlet	**243** °C	123 – 401
PIR 3.02	Ejector outlet	**2.3** mbar	1.8 – 9.5
Reformer			
TIR 3.06	Reformer inlet mixed	**112** °C	26.6 – 260
TIRC 3.07	Catalyst inlet	**124** °C	24.7 – 267
TIRC 3.15	Catalyst outlet	**133** °C	29.7 – 256
TIRC 3.17	Reformer outlet	**214** °C	101 – 308
Heater ERH			
TIRC 5.17	ERH control	**382** °C	255 – 506
TIRZ 8.2.5	ERH safety (trip 860)	**396** °C	266 – 523
IIR 5.1.3	ERH current	**3.2** A	0.00 – 4.7
Heater EAGH			
TIRC 3.1.1	EAGH control	**211** °C	145 – 337
TIRZ 8.2.1	EAGH safety (trip 950)	**213** °C	141 – 338
IIR 3.1.1	EAGH current	**3.0** A	0.00 – 4.8
TIR 3.18	Stack module anode inlet	**237** °C	107 – 362
PIR 3.03	Stack module anode inlet	**2.7** mbar	2.2 – 10.2
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**159** °C	33.7 – 274
TIRZ 8.1.3	Stack safety (trip 720)	**150** °C	32.2 – 275
TIR 3.19.1	Manifold 1 inlet	**95.5** °C	28.4 – 263

TIR 3.19.2	Manifold 2 inlet	**83.7** °C	26.1 – 270
TIRC 5.1.1	Cathode outlet 1	**143** °C	31.3 – 274
TIRC 5.1.2	Cathode outlet 2	**148** °C	32.4 – 275
PIR 5.1.1	Cathode outlet 1	**3.8** mbar	2.9 – 12.3
TIRC 5.2.1	Anode outlet 1	**158** °C	56.1 – 262
TIRC 5.2.2	Anode outlet 2	**118** °C	31.5 – 272
PIR 5.2.1	Anode outlet 1	**2.3** mbar	1.7 – 9.3
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.0
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.19** V	0.00 – 0.29
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**1.5** V	0.59 – 9.9
Σ stacks M2	String 2 stack-sum voltage	**1.9** V	0.46 – 12.8
Σ stacks M3	String 3 stack-sum voltage	**1.4** V	0.35 – 9.9
Σ stacks M4	String 4 stack-sum voltage	**2.0** V	0.41 – 14.2
Σ stacks M5	String 5 stack-sum voltage	**1.5** V	0.30 – 12.9
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**179** °C	129 – 273
TIRZ 8.2.4	EAOH3 safety (trip 950)	**177** °C	107 – 247

IIR 6.1.1	EAOH3 current	**8.2** A	0.00 – 10.9
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**273** °C	190 – 400
TIRZ 8.2.2	EAOH1 safety (trip 950)	**281** °C	191 – 413
IIR 5.1.1	EAOH1 current	**3.3** A	0.00 – 4.6
TIRC 5.5.1	TOX inlet anode 1	**320** °C	155 – 511
PIR 5.4.1	TOX inlet anode 1	**2.5** mbar	1.9 – 9.6
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**287** °C	203 – 419
TIRZ 8.2.3	EAOH2 safety (trip 950)	**294** °C	199 – 428
IIR 5.1.2	EAOH2 current	**3.1** A	0.00 – 4.6
TIRC 5.5.2	TOX inlet anode 2	**238** °C	86.3 – 338
PIR 5.4.2	TOX inlet anode 2	**2.6** mbar	2.1 – 9.8
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**142** °C	30.8 – 271
TIRZ 8.1.9	Burner safety (trip 930)	**142** °C	31.9 – 273
TIR 5.4.1	TOX inlet cathode 1	**141** °C	30.1 – 273
PIR 5.3.1	TOX inlet cathode 1	**3.6** mbar	2.9 – 11.7
TIR 5.6.1	Cathode bypass 1	**131** °C	33.0 – 261
TIR 7.1.1	TOX outlet 1	**142** °C	32.3 – 271
TIR 7.1.2	TOX outlet 2	**142** °C	33.7 – 270
Exhaust			
TIR 7.04	APH exhaust outlet	**27.0** °C	18.4 – 60.4
PIR 7.02	APH exhaust outlet	**0.32** mbar	0.29 – 0.63
TIR 7.05	Exhaust (roof)	**26.6** °C	18.1 – 59.9
PIR 7.03	Exhaust	**0.16** mbar	0.13 – 0.37

TIRZ 8.1.10	Exhaust safety (trip 300)	**28.3** °C	20.0 – 61.9

If you remember one thing: Heat-up air is the air heater and the gas-line heaters warming a cold stack to about 210 °C with air alone.

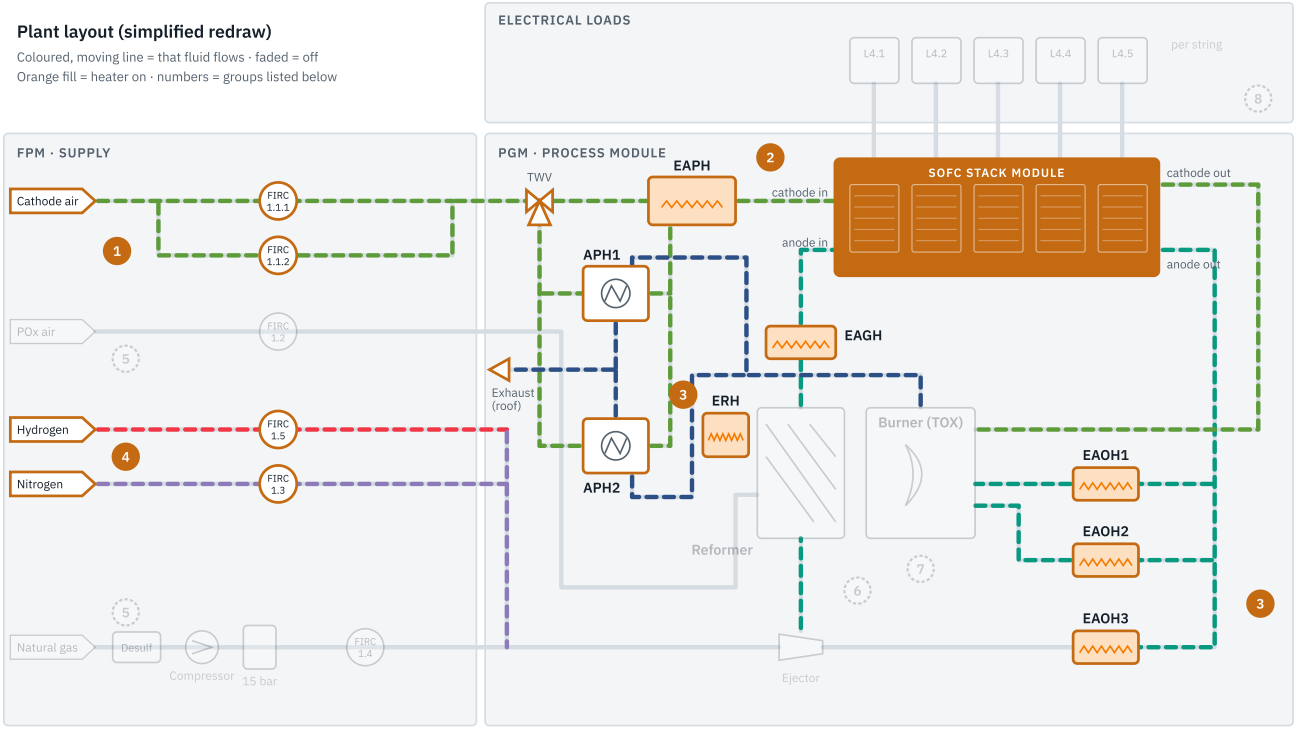
STAGE 2 OF 8 · START-UP WITH FORMING GAS

2 Heat-up N_2+H_2

> Once the stack passes about 210 °C, nitrogen with a little hydrogen is sent over the fuel side to protect it. The air heater now works hardest.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
15	81.5 h	7.2 h	9.8 h

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000

2 Electric air heater EAPH
On · 20 kW

3 Gas-line heaters ERH · EAGH · EAOH 1–3
On · 3 A

4 Forming gas N_2 + H_2
On

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

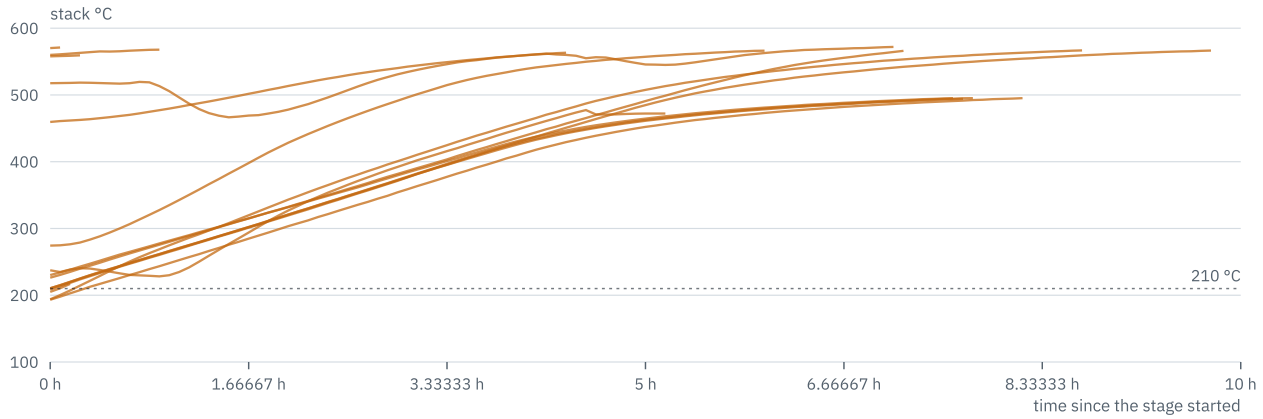
Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000
Electric air heater EAPH · 20 kW
Recuperators APH (passive)
Gas-line heaters ERH, EAGH, EAOH 1–2 · 3 A
Recycle heater EAOH 3 · 8 A
Forming gas N_2 1200 + H_2 60
~~Natural gas + reformer air~~
~~Reformer + gas recycle hot~~
~~After-burner burning~~
~~Stack current to loads~~
Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage



Every visit of this stage drawn from its start (one line per visit, 15 visits; shown up to 10.0 h). Stack keeps warming at about +39 °C/h. The voltage appears as soon as hydrogen reaches the cells, even with no current drawn. The after-burner is not burning yet.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**463** °C	265 – 557
Air into stack (TIRC 2.06)	**500** °C	346 – 571
Cathode air, both streams	**6,000** ln/min	5,999 – 9,999
Nitrogen N_2 (FIRC 1.3)	**1,200** ln/min	1,200 – 1,200
Hydrogen H_2 (FIRC 1.5)	**60.0** ln/min	60.0 – 60.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**419** V	212 – 459
Air heater EAPH power	**20.4** kW	13.3 – 28.6
Gas-line heater ERH current	**3.2** A	2.9 – 5.1
Recycle heater EAOH 3 current	**8.3** A	1.3 – 12.2
After-burner (TIRC 5.09)	**420** °C	272 – 518
Reformer catalyst inlet (TIRC 3.07)	**81.4** °C	51.6 – 124
Exhaust safety (TIRZ 8.1.10)	**87.8** °C	55.3 – 113

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. Typical visit ≈ 7 h. Next: Hot standby in 9 of 15 cases.

Heat-up air → Heat-up N_2+H_2 | Shutdown → Heat-up N_2+H_2 | Heat-up N_2+H_2 → Hot standby | Heat-up N_2+H_2 → Shutdown

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,000** ln/min	3,000 – 4,999
FIRC 1.1.2	Cathode air stream 2	**3,000** ln/min	3,000 – 4,999
PIR 1.1.1	Air stream 1 supply	**5.0** bar	4.5 – 5.8
PIR 1.1.3	Air stream 2 supply	**4.9** bar	4.3 – 5.8
TIR 1.1.1	Air after MFCs	**19.6** °C	18.0 – 26.6
PIR 1.1.2	Air after MFCs	**119** mbar	100 – 273
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**415** m³/h	251 – 484
TIR 2.04	TWV → APH inlet	**19.0** °C	18.2 – 26.9
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.5** bar	5.3 – 6.3
TIR 1.2.1	POx air (FPM)	**16.1** °C	11.6 – 18.2
TIR 1.2.2	POx air FPM → PGM	**6.0** °C	0.40 – 12.8
PIR 1.2.2	POx air after MFC	**64.3** mbar	61.0 – 73.3
TIR 3.04	POx air PGM inlet	**8.6** °C	3.9 – 14.6
TIR 3.05	Reformer inlet POx air	**10.3** °C	5.5 – 17.4
Nitrogen			
FIRC 1.3	Nitrogen	**1,200** ln/min	1,200 – 1,200
PIR 1.3.1	N_2 supply	**4.7** bar	4.6 – 4.7
TIR 1.3	Purge gas	**5.6** °C	0.80 – 12.4
PIR 1.3.2	Purge gas	**24.0** bar?	23.5 – 24.3
Hydrogen			
FIRC 1.5	Hydrogen	**60.0** ln/min	60.0 – 60.0
PIR 1.5	H_2 supply	**4.9** bar	4.7 – 5.3

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**0.04** bar	0.02 – 13.7
TIR 1.4.1	Fuel (FPM)	**16.4** °C	11.4 – 18.3
PIR 1.4.2	Fuel after MFC	**0.02** bar	0.01 – 0.03
TIR 1.4.2	Fuel FPM → PGM	**6.5** °C	0.90 – 13.4
TIR 3.01	Fuel PGM inlet	**13.5** °C	6.2 – 27.4
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**18.5** °C	17.5 – 27.0
TIR 2.02	APH1+2 outlet air	**367** °C	221 – 475
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**391** °C	235 – 502
PIR 7.1.1	APH inlet exhaust 1	**4.0** mbar	3.0 – 8.0
TIR 7.3.1	Flue gas out 1	**71.8** °C	44.2 – 95.3
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**18.3** °C	17.3 – 26.8
TIR 2.03	APH outlet riser → EAPH	**366** °C	220 – 475
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**395** °C	240 – 507
PIR 7.1.2	APH inlet exhaust 2	**4.1** mbar	2.9 – 8.5
TIR 7.3.2	Flue gas out 2	**72.3** °C	44.3 – 94.1
Air heater EAPH			
TIR 2.05	EAPH inlet	**367** °C	221 – 475
PIR 2.01	EAPH inlet	**58.8** mbar	43.3 – 129
EAPH power	Heater power (3-phase)	**20.4** kW	13.3 – 28.6
TIRZ 8.1.1	EAPH safety (trip 1050)	**510** °C	342 – 588
TIRC 2.06	EAPH outlet = stack air in	**501** °C	346 – 571
PIR 2.02	Stack module air inlet	**11.4** mbar	8.5 – 23.4

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**20.6** °C	12.7 – 41.3
PIR 3.01	Ejector main stream	**0.02** bar	0.01 – 0.03
TIR 6.01	Recycle gas inlet	**71.9** °C	52.6 – 104
PIR 6.01	Recycle gas inlet	**13.4** mbar	11.2 – 22.5
TIRC 3.03	Ejector outlet	**16.8** °C	10.7 – 26.3
PIR 3.02	Ejector outlet	**16.7** mbar	13.6 – 26.0
Reformer			
TIR 3.06	Reformer inlet mixed	**15.2** °C	9.8 – 25.4
TIRC 3.07	Catalyst inlet	**81.4** °C	51.6 – 124
TIRC 3.15	Catalyst outlet	**349** °C	215 – 469
TIRC 3.17	Reformer outlet	**372** °C	231 – 495
Heater ERH			
TIRC 5.17	ERH control	**559** °C	450 – 729
TIRZ 8.2.5	ERH safety (trip 860)	**572** °C	461 – 751
IIR 5.1.3	ERH current	**3.2** A	2.9 – 5.1
Heater EAGH			
TIRC 3.1.1	EAGH control	**377** °C	261 – 518
TIRZ 8.2.1	EAGH safety (trip 950)	**382** °C	262 – 520
IIR 3.1.1	EAGH current	**3.2** A	2.8 – 4.9
TIR 3.18	Stack module anode inlet	**376** °C	247 – 519
PIR 3.03	Stack module anode inlet	**15.0** mbar	12.6 – 24.1
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**463** °C	265 – 557
TIRZ 8.1.3	Stack safety (trip 720)	**457** °C	254 – 556
TIR 3.19.1	Manifold 1 inlet	**360** °C	235 – 496

TIR 3.19.2	Manifold 2 inlet	**352** °C	226 – 491
TIRC 5.1.1	Cathode outlet 1	**449** °C	248 – 552
TIRC 5.1.2	Cathode outlet 2	**455** °C	258 – 555
PIR 5.1.1	Cathode outlet 1	**9.8** mbar	7.4 – 21.0
TIRC 5.2.1	Anode outlet 1	**409** °C	238 – 540
TIRC 5.2.2	Anode outlet 2	**427** °C	228 – 543
PIR 5.2.1	Anode outlet 1	**13.1** mbar	10.9 – 22.2
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.0
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.17** V	0.0 – 0.24
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**419** V	212 – 459
Σ stacks M2	String 2 stack-sum voltage	**436** V	310 – 468
Σ stacks M3	String 3 stack-sum voltage	**412** V	337 – 461
Σ stacks M4	String 4 stack-sum voltage	**420** V	318 – 456
Σ stacks M5	String 5 stack-sum voltage	**469** V	367 – 486
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**144** °C	112 – 226
TIRZ 8.2.4	EAOH3 safety (trip 950)	**122** °C	95.0 – 181

IIR 6.1.1	EAOH3 current	**8.3** A	1.3 – 12.2
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**429** °C	314 – 557
TIRZ 8.2.2	EAOH1 safety (trip 950)	**430** °C	313 – 560
IIR 5.1.1	EAOH1 current	**3.4** A	2.9 – 5.1
TIRC 5.5.1	TOX inlet anode 1	**303** °C	231 – 392
PIR 5.4.1	TOX inlet anode 1	**12.4** mbar	10.3 – 21.3
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**443** °C	323 – 579
TIRZ 8.2.3	EAOH2 safety (trip 950)	**442** °C	318 – 580
IIR 5.1.2	EAOH2 current	**3.2** A	2.8 – 5.0
TIRC 5.5.2	TOX inlet anode 2	**312** °C	235 – 412
PIR 5.4.2	TOX inlet anode 2	**12.5** mbar	10.4 – 21.5
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**420** °C	272 – 518
TIRZ 8.1.9	Burner safety (trip 930)	**404** °C	257 – 510
TIR 5.4.1	TOX inlet cathode 1	**445** °C	247 – 548
PIR 5.3.1	TOX inlet cathode 1	**9.5** mbar	7.2 – 19.9
TIR 5.6.1	Cathode bypass 1	**374** °C	223 – 487
TIR 7.1.1	TOX outlet 1	**410** °C	241 – 521
TIR 7.1.2	TOX outlet 2	**401** °C	244 – 512
Exhaust			
TIR 7.04	APH exhaust outlet	**87.9** °C	54.6 – 113
PIR 7.02	APH exhaust outlet	**0.40** mbar	0.36 – 0.81
TIR 7.05	Exhaust (roof)	**86.8** °C	53.9 – 112
PIR 7.03	Exhaust	**0.22** mbar	0.19 – 0.47

TIRZ 8.1.10	Exhaust safety (trip 300)	**87.8** °C	55.3 – 113

If you remember one thing: Heat-up N_2+H_2 is the same warming with protective forming gas on, and it is where the stack voltage first appears.

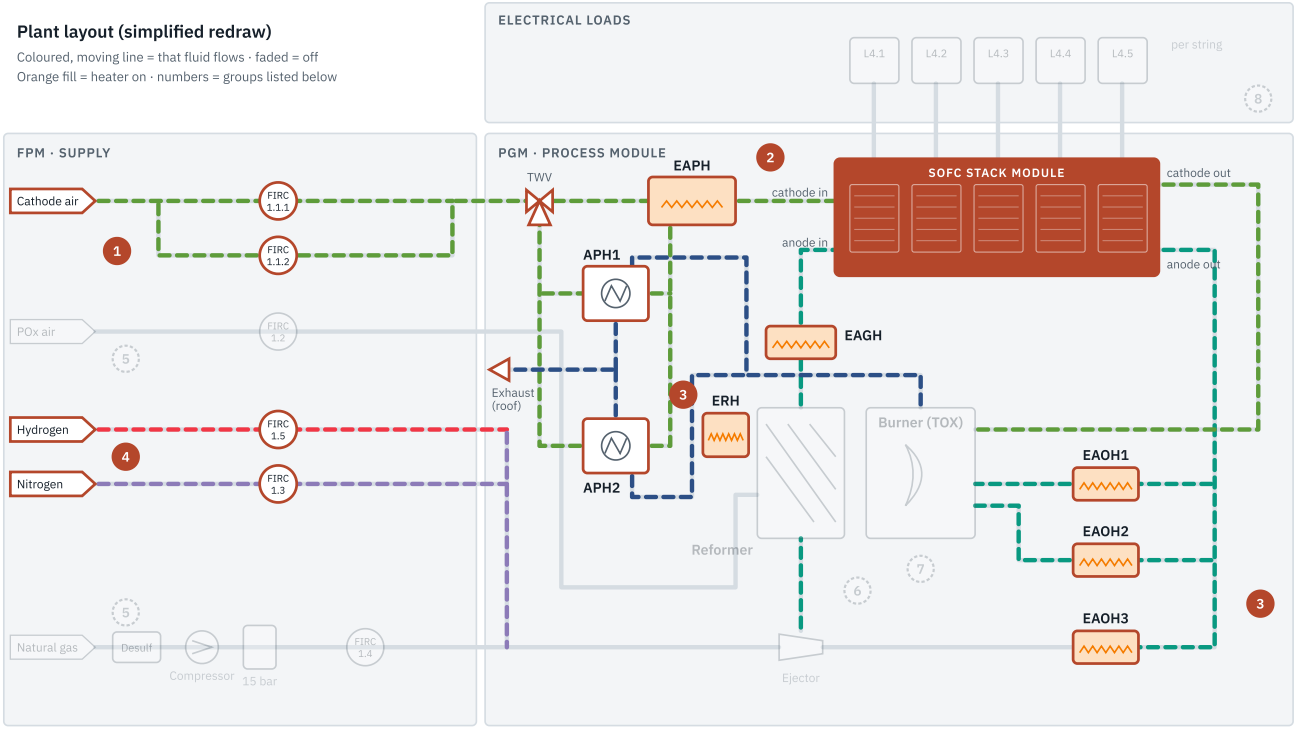
STAGE 3 OF 8 · HOT AND READY, NO POWER

3 Hot standby

The stack is held hot with forming gas on, waiting for the order to make power. This is where the plant spent most of its time.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
28	338 h	2.8 h	80.2 h

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000

2 Electric air heater EAPH
On · 10 kW

3 Gas-line heaters ERH · EAGH · EAOH 1–3
On · 3 A

4 Forming gas N_2 + H_2
On

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

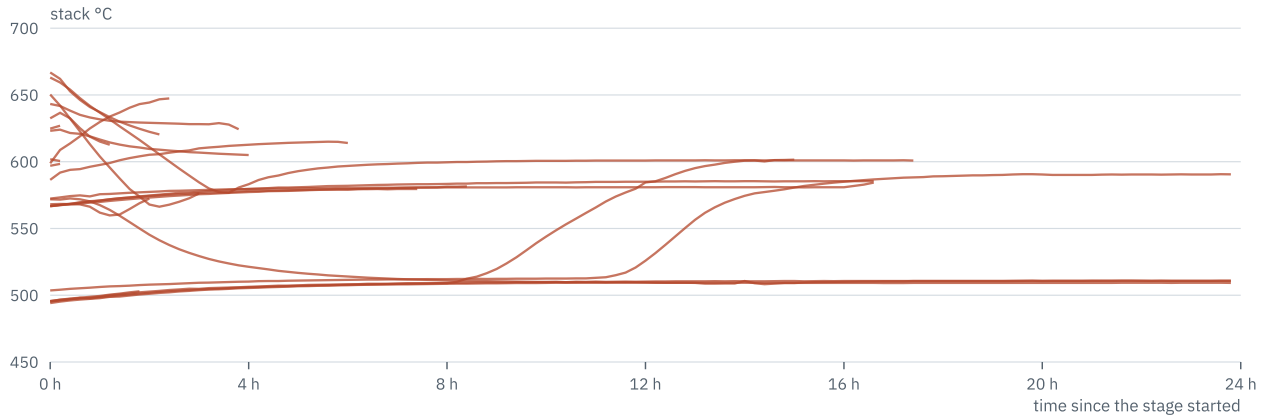
Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000
Electric air heater EAPH · 10 kW
Recuperators APH (passive)
Gas-line heaters ERH, EAGH, EAOH 1–2 · 3 A
Recycle heater EAOH 3 · 8 A
Forming gas N_2 1200 + H_2 60
~~Natural gas + reformer air~~
~~Reformer + gas recycle hot~~
~~After-burner burning~~
~~Stack current to loads~~
Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage



Every visit of this stage drawn from its start (one line per visit, 28 visits; shown up to 24.0 h). Readings stay flat. Re-heats and cool-backs (the old “thermal adjustment”) are started by operator commands. The compressor fills its tank near the end.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**511** °C	509 – 600
Air into stack (TIRC 2.06)	**500** °C	500 – 581
Cathode air, both streams	**6,000** ln/min	5,999 – 9,999
Nitrogen N_2 (FIRC 1.3)	**1,200** ln/min	1,200 – 1,200
Hydrogen H_2 (FIRC 1.5)	**60.0** ln/min	60.0 – 60.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**429** V	226 – 438
Air heater EAPH power	**9.6** kW	8.8 – 16.0
Gas-line heater ERH current	**3.2** A	3.0 – 5.2
Recycle heater EAOH 3 current	**8.4** A	7.9 – 12.9
After-burner (TIRC 5.09)	**478** °C	471 – 558
Reformer catalyst inlet (TIRC 3.07)	**101** °C	96.2 – 147
Exhaust safety (TIRZ 8.1.10)	**109** °C	106 – 128

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. 338 h in total; visits from minutes to 80 h. Next: Fuel switch (17 of 25) or Air-flow cooling (6).

Heat-up N_2+H_2 → Hot standby | Shutdown → Hot standby | Hot standby → Fuel switch

Fuel switch → Hot standby | Part load → Hot standby | Hot standby → Air-flow cooling

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,000** ln/min	3,000 – 5,000
FIRC 1.1.2	Cathode air stream 2	**3,000** ln/min	3,000 – 5,000
PIR 1.1.1	Air stream 1 supply	**5.0** bar	4.7 – 5.2
PIR 1.1.3	Air stream 2 supply	**4.9** bar	4.6 – 5.2
TIR 1.1.1	Air after MFCs	**19.3** °C	17.4 – 20.9
PIR 1.1.2	Air after MFCs	**122** mbar	120 – 286
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**513** m^3/h	439 – 543
TIR 2.04	TWV → APH inlet	**18.8** °C	17.6 – 22.3
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.3** bar	5.3 – 6.2
TIR 1.2.1	POx air (FPM)	**15.8** °C	11.4 – 17.3
TIR 1.2.2	POx air FPM → PGM	**3.9** °C	0.0 – 12.7
PIR 1.2.2	POx air after MFC	**66.7** mbar	65.1 – 77.1
TIR 3.04	POx air PGM inlet	**4.8** °C	1.6 – 13.0
TIR 3.05	Reformer inlet POx air	**7.3** °C	4.1 – 16.5
Nitrogen			
FIRC 1.3	Nitrogen	**1,200** ln/min	1,200 – 1,200
PIR 1.3.1	N_2 supply	**4.7** bar	4.6 – 4.8
TIR 1.3	Purge gas	**3.0** °C	0.07 – 11.1
PIR 1.3.2	Purge gas	**24.1** bar?	23.9 – 24.4
Hydrogen			
FIRC 1.5	Hydrogen	**60.0** ln/min	60.0 – 60.0
PIR 1.5	H_2 supply	**5.0** bar	4.7 – 5.3

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**0.83** bar	0.02 – 13.7
TIR 1.4.1	Fuel (FPM)	**15.6** °C	10.9 – 17.0
PIR 1.4.2	Fuel after MFC	**0.02** bar	0.02 – 0.03
TIR 1.4.2	Fuel FPM → PGM	**4.0** °C	0.29 – 13.2
TIR 3.01	Fuel PGM inlet	**15.1** °C	10.0 – 28.3
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**18.3** °C	17.1 – 22.4
TIR 2.02	APH1+2 outlet air	**440** °C	433 – 520
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**460** °C	454 – 551
PIR 7.1.1	APH inlet exhaust 1	**4.4** mbar	4.3 – 9.3
TIR 7.3.1	Flue gas out 1	**89.6** °C	87.6 – 105
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**18.1** °C	16.8 – 22.2
TIR 2.03	APH outlet riser → EAPH	**440** °C	433 – 521
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**462** °C	455 – 560
PIR 7.1.2	APH inlet exhaust 2	**5.0** mbar	4.2 – 9.3
TIR 7.3.2	Flue gas out 2	**89.4** °C	87.3 – 106
Air heater EAPH			
TIR 2.05	EAPH inlet	**440** °C	434 – 521
PIR 2.01	EAPH inlet	**60.2** mbar	59.1 – 140
EAPH power	Heater power (3-phase)	**9.6** kW	8.8 – 16.0
TIRZ 8.1.1	EAPH safety (trip 1050)	**504** °C	502 – 586
TIRC 2.06	EAPH outlet = stack air in	**500** °C	500 – 581
PIR 2.02	Stack module air inlet	**12.4** mbar	12.2 – 27.0

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**29.3** °C	18.9 – 39.0
PIR 3.01	Ejector main stream	**0.02** bar	0.02 – 0.03
TIR 6.01	Recycle gas inlet	**51.6** °C	48.3 – 96.9
PIR 6.01	Recycle gas inlet	**15.1** mbar	14.6 – 25.8
TIRC 3.03	Ejector outlet	**12.3** °C	8.9 – 23.2
PIR 3.02	Ejector outlet	**18.6** mbar	18.0 – 30.0
Reformer			
TIR 3.06	Reformer inlet mixed	**14.8** °C	11.3 – 27.3
TIRC 3.07	Catalyst inlet	**101** °C	96.2 – 147
TIRC 3.15	Catalyst outlet	**420** °C	414 – 531
TIRC 3.17	Reformer outlet	**447** °C	439 – 556
Heater ERH			
TIRC 5.17	ERH control	**623** °C	614 – 781
TIRZ 8.2.5	ERH safety (trip 860)	**636** °C	626 – 802
IIR 5.1.3	ERH current	**3.2** A	3.0 – 5.2
Heater EAGH			
TIRC 3.1.1	EAGH control	**469** °C	459 – 593
TIRZ 8.2.1	EAGH safety (trip 950)	**471** °C	461 – 593
IIR 3.1.1	EAGH current	**2.9** A	2.8 – 4.8
TIR 3.18	Stack module anode inlet	**462** °C	453 – 591
PIR 3.03	Stack module anode inlet	**16.8** mbar	16.2 – 27.7
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**511** °C	509 – 600
TIRZ 8.1.3	Stack safety (trip 720)	**514** °C	512 – 605
TIR 3.19.1	Manifold 1 inlet	**455** °C	445 – 570

TIR 3.19.2	Manifold 2 inlet	**454** °C	445 – 567
TIRC 5.1.1	Cathode outlet 1	**509** °C	507 – 597
TIRC 5.1.2	Cathode outlet 2	**512** °C	510 – 602
PIR 5.1.1	Cathode outlet 1	**10.7** mbar	10.5 – 24.0
TIRC 5.2.1	Anode outlet 1	**508** °C	502 – 610
TIRC 5.2.2	Anode outlet 2	**507** °C	503 – 623
PIR 5.2.1	Anode outlet 1	**14.7** mbar	14.2 – 25.5
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.0
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.19** V	0.0 – 0.29
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**429** V	226 – 438
Σ stacks M2	String 2 stack-sum voltage	**439** V	326 – 447
Σ stacks M3	String 3 stack-sum voltage	**418** V	407 – 432
Σ stacks M4	String 4 stack-sum voltage	**432** V	411 – 438
Σ stacks M5	String 5 stack-sum voltage	**469** V	450 – 479
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**114** °C	110 – 219
TIRZ 8.2.4	EAOH3 safety (trip 950)	**93.9** °C	86.6 – 172

IIR 6.1.1	EAOH3 current	**8.4** A	7.9 – 12.9
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**500** °C	490 – 608
TIRZ 8.2.2	EAOH1 safety (trip 950)	**502** °C	494 – 611
IIR 5.1.1	EAOH1 current	**3.3** A	3.0 – 5.1
TIRC 5.5.1	TOX inlet anode 1	**338** °C	332 – 428
PIR 5.4.1	TOX inlet anode 1	**13.9** mbar	13.4 – 24.5
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**511** °C	503 – 656
TIRZ 8.2.3	EAOH2 safety (trip 950)	**516** °C	509 – 661
IIR 5.1.2	EAOH2 current	**3.1** A	2.9 – 5.1
TIRC 5.5.2	TOX inlet anode 2	**347** °C	340 – 474
PIR 5.4.2	TOX inlet anode 2	**14.1** mbar	13.6 – 24.6
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**478** °C	471 – 558
TIRZ 8.1.9	Burner safety (trip 930)	**459** °C	451 – 556
TIR 5.4.1	TOX inlet cathode 1	**507** °C	504 – 594
PIR 5.3.1	TOX inlet cathode 1	**10.3** mbar	10.1 – 22.8
TIR 5.6.1	Cathode bypass 1	**455** °C	447 – 541
TIR 7.1.1	TOX outlet 1	**478** °C	473 – 568
TIR 7.1.2	TOX outlet 2	**464** °C	457 – 563
Exhaust			
TIR 7.04	APH exhaust outlet	**108** °C	106 – 127
PIR 7.02	APH exhaust outlet	**0.39** mbar	0.35 – 0.97
TIR 7.05	Exhaust (roof)	**107** °C	105 – 126
PIR 7.03	Exhaust	**0.25** mbar	0.21 – 0.57

TIRZ 8.1.10	Exhaust safety (trip 300)	**109** °C	107 – 128

If you remember one thing: Hot standby is a hot stack held flat with forming gas on, waiting; the plant spent more time here than anywhere else.

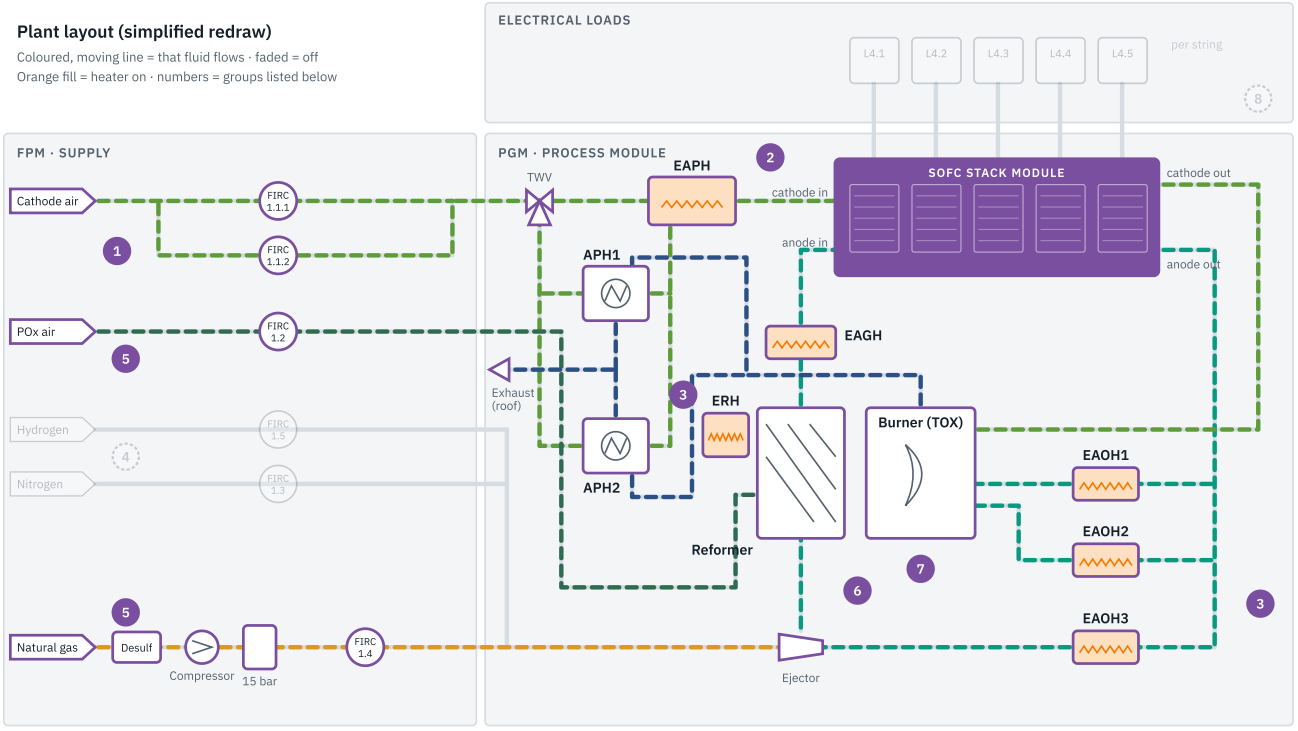
STAGE 4 OF 8 · NATURAL GAS ON, NO CURRENT YET

4 Fuel switch

Forming gas goes off and natural gas plus reformer air come on. The reformer and the after-burner light up. The stack is ready but no current is drawn yet.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
19	3.3 h	8 min	29 min

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000–8000

2 Electric air heater EAPH
On · 11 kW

3 Gas-line heaters ERH · EAGH · EAOH 1–3
On · 5 A

4 Forming gas N_2 + H_2
Off

5 Natural gas + reformer air (POx)
On · 80 + 343

6 Reformer + gas recycle (ejector)
On

7 After-burner (TOX) burning
Partly · starts burning as the gas comes in

8 Stack current to the loads (power out)
Off

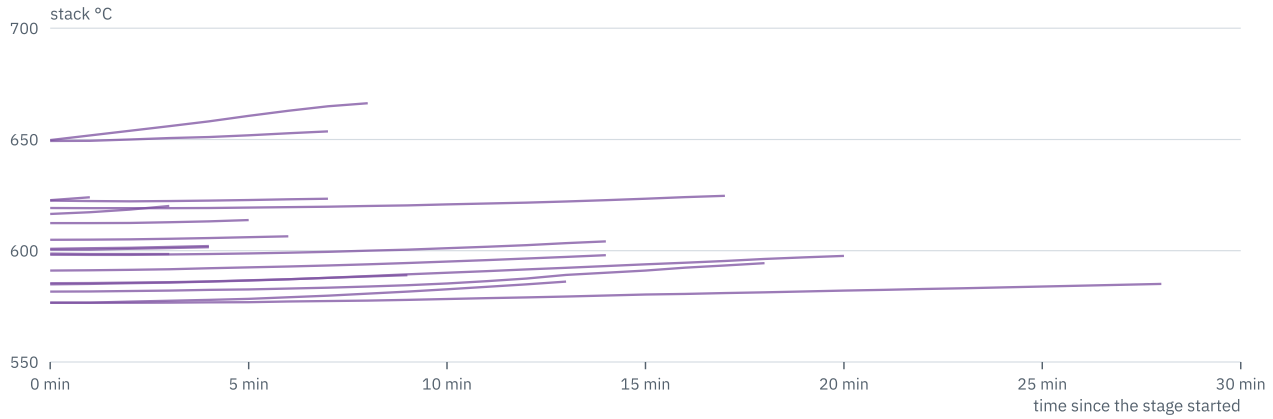
· Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000–8000 | Electric air heater EAPH · 11 kW | Recuperators APH (passive) | Gas-line heaters ERH, EAGH, EAOH 1–2 · 5 A | Recycle heater EAOH 3 · 12 A | ~~Forming gas N_2 1200 + H_2 60~~ | Natural gas + reformer air · 80 + 343 | Reformer + gas recycle hot | After-burner burning · starting (partly) | ~~Stack current to loads · < 1 A~~ | Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

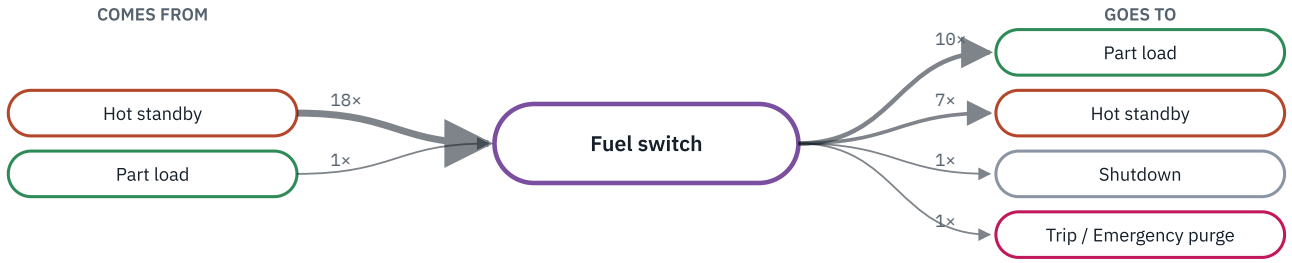


Every visit of this stage drawn from its start (one line per visit, 19 visits; shown up to 30 min). The stack never settles here. Gas heat pushes it up while the voltage climbs to its open-circuit value. Gas-line heaters step up to ≈ 5 A, recycle heater ≈ 12 A.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**595** °C	579 – 624
Air into stack (TIRC 2.06)	**581** °C	572 – 600
Cathode air, both streams	**7,999** ln/min	5,999 – 9,990
Nitrogen N_2 (FIRC 1.3)	**0.0** ln/min	0.0 – 120
Hydrogen H_2 (FIRC 1.5)	**0.0** ln/min	0.0 – 6.0
Natural gas (FIRC 1.4)	**80.0** ln/min	46.7 – 140
Reformer air (FIRC 1.2)	**349** ln/min	197 – 557
String 1 current	**0.0** A	0.0 – 0.30
String 1 stack voltage	**460** V	426 – 474
Air heater EAPH power	**11.4** kW	5.2 – 17.3
Gas-line heater ERH current	**5.2** A	3.1 – 5.3
Recycle heater EAOH 3 current	**12.1** A	1.8 – 13.1
After-burner (TIRC 5.09)	**647** °C	575 – 779
Reformer catalyst inlet (TIRC 3.07)	**762** °C	435 – 808
Exhaust safety (TIRZ 8.1.10)	**114** °C	96.4 – 141

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. Typical visit ≈ 8 min (max 29). Gas-on to first current: median 10 min. Next: Part load (10 of 19) or back to Hot standby (7).

Hot standby → Fuel switch | Fuel switch → Part load | Fuel switch → Hot standby

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,002** ln/min	3,000 – 4,994
FIRC 1.1.2	Cathode air stream 2	**3,002** ln/min	3,000 – 4,996
PIR 1.1.1	Air stream 1 supply	**4.9** bar	3.8 – 5.7
PIR 1.1.3	Air stream 2 supply	**4.8** bar	3.7 – 5.7
TIR 1.1.1	Air after MFCs	**17.8** °C	16.4 – 23.1
PIR 1.1.2	Air after MFCs	**137** mbar	125 – 283
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**539** m³/h	516 – 589
TIR 2.04	TWV → APH inlet	**19.6** °C	17.9 – 24.5
Reformer air (POx)			
FIRC 1.2	POx air	**343** ln/min	134 – 556
PIR 1.2.1	POx air supply	**5.2** bar	4.2 – 6.0
TIR 1.2.1	POx air (FPM)	**16.1** °C	12.5 – 19.6
TIR 1.2.2	POx air FPM → PGM	**15.0** °C	11.1 – 17.7
PIR 1.2.2	POx air after MFC	**86.7** mbar	34.4 – 164
TIR 3.04	POx air PGM inlet	**13.3** °C	9.6 – 19.0
TIR 3.05	Reformer inlet POx air	**33.8** °C	26.3 – 49.9
Nitrogen			
FIRC 1.3	Nitrogen	**0.0** ln/min	0.0 – 161
PIR 1.3.1	N_2 supply	**5.5** bar	5.3 – 5.5
TIR 1.3	Purge gas	**9.3** °C	5.4 – 13.4
PIR 1.3.2	Purge gas	**1.7** bar?	1.1 – 4.2
Hydrogen			
FIRC 1.5	Hydrogen	**0.0** ln/min	0.0 – 8.0
PIR 1.5	H_2 supply	**5.4** bar	5.2 – 5.5

Natural gas			
FIRC 1.4	Natural gas	**80.0** ln/min	33.7 – 140
PIR 1.4.1	After booster	**13.4** bar	13.4 – 13.6
TIR 1.4.1	Fuel (FPM)	**13.1** °C	7.7 – 16.1
PIR 1.4.2	Fuel after MFC	**5.2** bar	1.6 – 9.0
TIR 1.4.2	Fuel FPM → PGM	**15.6** °C	12.8 – 18.0
TIR 3.01	Fuel PGM inlet	**15.7** °C	11.2 – 19.7
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**19.6** °C	17.9 – 24.6
TIR 2.02	APH1+2 outlet air	**536** °C	506 – 589
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**591** °C	550 – 658
PIR 7.1.1	APH inlet exhaust 1	**5.7** mbar	4.1 – 8.5
TIR 7.3.1	Flue gas out 1	**95.6** °C	81.3 – 142
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**19.3** °C	17.6 – 24.3
TIR 2.03	APH outlet riser → EAPH	**535** °C	505 – 587
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**597** °C	557 – 668
PIR 7.1.2	APH inlet exhaust 2	**5.6** mbar	4.0 – 8.4
TIR 7.3.2	Flue gas out 2	**94.9** °C	79.6 – 140
Air heater EAPH			
TIR 2.05	EAPH inlet	**535** °C	505 – 581
PIR 2.01	EAPH inlet	**73.6** mbar	63.1 – 139
EAPH power	Heater power (3-phase)	**11.4** kW	5.3 – 17.3
TIRZ 8.1.1	EAPH safety (trip 1050)	**582** °C	554 – 603
TIRC 2.06	EAPH outlet = stack air in	**581** °C	571 – 600
PIR 2.02	Stack module air inlet	**14.6** mbar	10.6 – 24.2

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**22.5** °C	18.8 – 28.5
PIR 3.01	Ejector main stream	**5.1** bar	1.7 – 9.0
TIR 6.01	Recycle gas inlet	**529** °C	348 – 629
PIR 6.01	Recycle gas inlet	**14.3** mbar	7.5 – 19.0
TIRC 3.03	Ejector outlet	**345** °C	175 – 438
PIR 3.02	Ejector outlet	**27.7** mbar	12.9 – 33.7
Reformer			
TIR 3.06	Reformer inlet mixed	**267** °C	127 – 396
TIRC 3.07	Catalyst inlet	**756** °C	415 – 808
TIRC 3.15	Catalyst outlet	**550** °C	502 – 635
TIRC 3.17	Reformer outlet	**566** °C	527 – 632
Heater ERH			
TIRC 5.17	ERH control	**786** °C	650 – 804
TIRZ 8.2.5	ERH safety (trip 860)	**807** °C	661 – 826
IIR 5.1.3	ERH current	**5.2** A	3.1 – 5.3
Heater EAGH			
TIRC 3.1.1	EAGH control	**583** °C	555 – 618
TIRZ 8.2.1	EAGH safety (trip 950)	**583** °C	557 – 620
IIR 3.1.1	EAGH current	**4.8** A	0.12 – 4.9
TIR 3.18	Stack module anode inlet	**584** °C	552 – 623
PIR 3.03	Stack module anode inlet	**18.8** mbar	9.1 – 24.3
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**595** °C	580 – 623
TIRZ 8.1.3	Stack safety (trip 720)	**603** °C	583 – 635
TIR 3.19.1	Manifold 1 inlet	**574** °C	535 – 594

TIR 3.19.2	Manifold 2 inlet	**572** °C	530 – 592
TIRC 5.1.1	Cathode outlet 1	**594** °C	577 – 623
TIRC 5.1.2	Cathode outlet 2	**597** °C	581 – 627
PIR 5.1.1	Cathode outlet 1	**12.9** mbar	8.9 – 21.2
TIRC 5.2.1	Anode outlet 1	**621** °C	586 – 667
TIRC 5.2.2	Anode outlet 2	**608** °C	577 – 662
PIR 5.2.1	Anode outlet 1	**14.3** mbar	7.3 – 19.0
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.25
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.15
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.40
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.19
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.16
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 466
EIRC 4.2.1	String 2 load voltage	**0.23** V	0.10 – 470
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 446
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 465
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 484
Σ stacks M1	String 1 stack-sum voltage	**458** V	413 – 473
Σ stacks M2	String 2 stack-sum voltage	**461** V	418 – 476
Σ stacks M3	String 3 stack-sum voltage	**452** V	404 – 473
Σ stacks M4	String 4 stack-sum voltage	**456** V	414 – 473
Σ stacks M5	String 5 stack-sum voltage	**471** V	449 – 484
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**347** °C	220 – 488
TIRZ 8.2.4	EAOH3 safety (trip 950)	**310** °C	183 – 448

IIR 6.1.1	EAOH3 current	**12.1** A	2.3 – 13.1
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**630** °C	568 – 671
TIRZ 8.2.2	EAOH1 safety (trip 950)	**631** °C	575 – 671
IIR 5.1.1	EAOH1 current	**4.6** A	0.10 – 5.1
TIRC 5.5.1	TOX inlet anode 1	**572** °C	491 – 665
PIR 5.4.1	TOX inlet anode 1	**13.9** mbar	7.2 – 18.6
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**643** °C	587 – 688
TIRZ 8.2.3	EAOH2 safety (trip 950)	**646** °C	593 – 695
IIR 5.1.2	EAOH2 current	**4.5** A	0.12 – 5.2
TIRC 5.5.2	TOX inlet anode 2	**584** °C	510 – 676
PIR 5.4.2	TOX inlet anode 2	**14.1** mbar	7.4 – 18.7
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**643** °C	574 – 778
TIRZ 8.1.9	Burner safety (trip 930)	**626** °C	571 – 763
TIR 5.4.1	TOX inlet cathode 1	**592** °C	575 – 622
PIR 5.3.1	TOX inlet cathode 1	**12.3** mbar	8.5 – 20.1
TIR 5.6.1	Cathode bypass 1	**564** °C	530 – 611
TIR 7.1.1	TOX outlet 1	**597** °C	562 – 669
TIR 7.1.2	TOX outlet 2	**611** °C	561 – 699
Exhaust			
TIR 7.04	APH exhaust outlet	**113** °C	95.2 – 168
PIR 7.02	APH exhaust outlet	**0.59** mbar	0.42 – 0.86
TIR 7.05	Exhaust (roof)	**113** °C	95.0 – 164
PIR 7.03	Exhaust	**0.35** mbar	0.26 – 0.51

TIRZ 8.1.10	Exhaust safety (trip 300)	**114** °C	96.5 – 165

If you remember one thing: **Fuel switch is the few minutes when natural gas is on and the reformer lights up, but no current flows yet.**

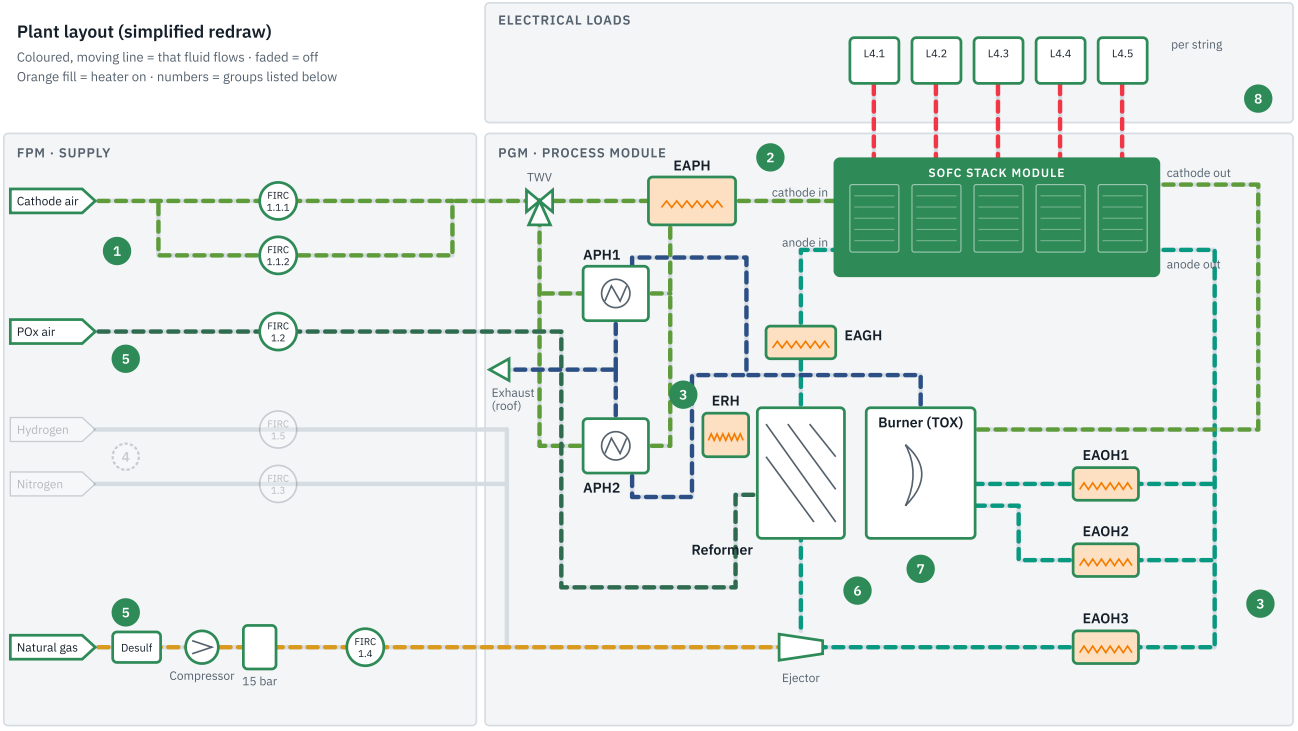
STAGE 5 OF 8 · MAKING POWER

5 Part load

Current flows from the five stack strings into the loads. More natural gas and more air are needed. Full rated power was never reached.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
10	4.5 h	28 min	44 min

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 9000

2 Electric air heater EAPH
On · 9 kW

3 Gas-line heaters ERH · EAGH · EAOH 1–3
On · 3–5 A

4 Forming gas N_2 + H_2
Off

5 Natural gas + reformer air (POx)
On · 137 + 312

6 Reformer + gas recycle (ejector)
On

7 After-burner (TOX) burning
On

8 Stack current to the loads (power out)
On · 6–7 A

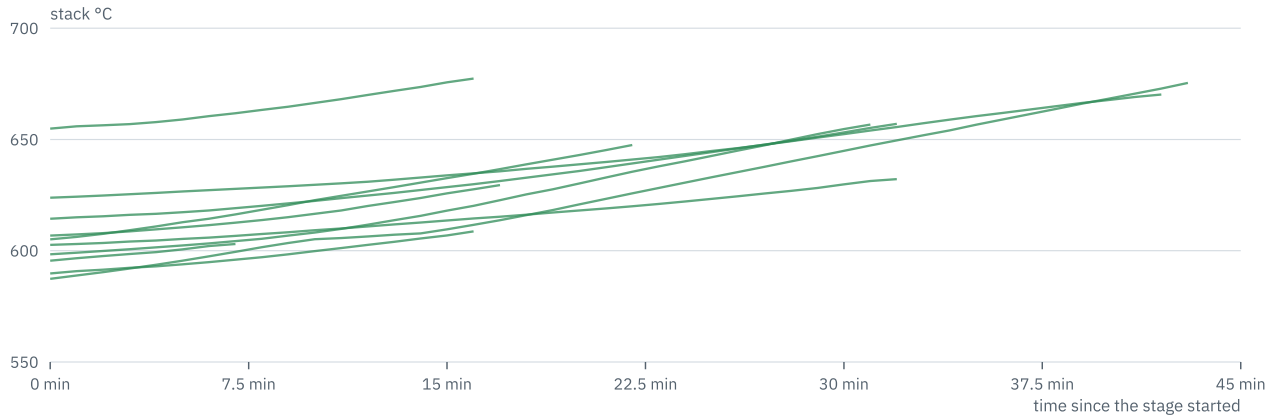
· Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 9000 | Electric air heater EAPH · 9 kW | Recuperators APH (passive) | Gas-line heaters ERH, EAGH, EAOH 1–2 · 3–5 A | Recycle heater EAOH 3 · 4 A | ~~Forming gas N_2 1200 + H_2 60~~ | Natural gas + reformer air · 137 + 312 | Reformer + gas recycle hot | After-burner burning | Stack current to loads · 6–7 A | Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

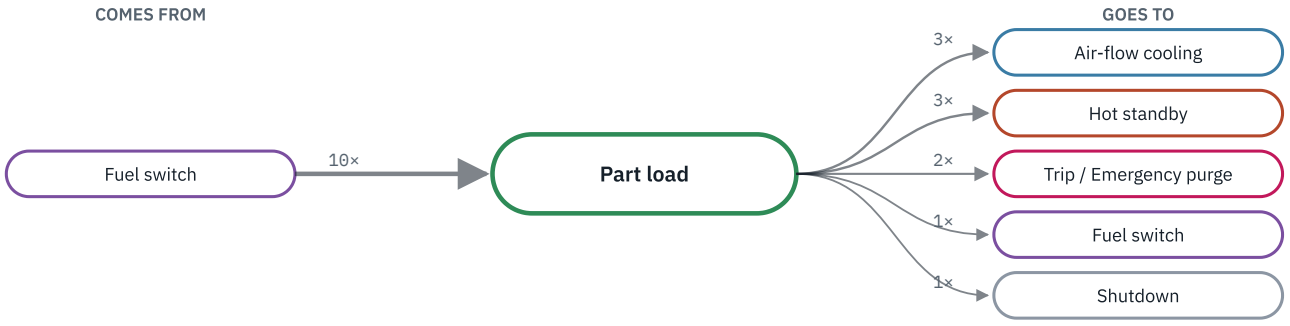


Every visit of this stage drawn from its start (one line per visit, 10 visits; shown up to 45 min). The stack heats fast because the current itself makes heat. No run lasted long enough to settle; at 720 °C the plant trips.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**625** °C	600 – 659
Air into stack (TIRC 2.06)	**600** °C	580 – 610
Cathode air, both streams	**9,000** ln/min	6,000 – 9,984
Nitrogen N_2 (FIRC 1.3)	**0.0** ln/min	0.0 – 0.0
Hydrogen H_2 (FIRC 1.5)	**0.0** ln/min	0.0 – 0.0
Natural gas (FIRC 1.4)	**137** ln/min	100 – 150
Reformer air (FIRC 1.2)	**312** ln/min	240 – 491
String 1 current	**6.5** A	2.0 – 14.0
String 1 stack voltage	**430** V	390 – 458
Air heater EAPH power	**8.6** kW	3.3 – 18.6
Gas-line heater ERH current	**5.1** A	3.0 – 5.2
Recycle heater EAOH 3 current	**4.2** A	0.10 – 10.0
After-burner (TIRC 5.09)	**741** °C	678 – 790
Reformer catalyst inlet (TIRC 3.07)	**712** °C	681 – 809
Exhaust safety (TIRZ 8.1.10)	**114** °C	107 – 183

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. 10 runs, 4.5 h in total, longest 44 min. Next: Air-flow cooling (3), Hot standby (2), Fuel switch while unloading (2), Trip (2).

Fuel switch → Part load | Part load → Hot standby | Part load → Air-flow cooling

Part load → Trip / Emergency purge

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**4,500** ln/min	3,000 – 4,976
FIRC 1.1.2	Cathode air stream 2	**4,500** ln/min	3,000 – 4,991
PIR 1.1.1	Air stream 1 supply	**5.1** bar	3.8 – 5.6
PIR 1.1.3	Air stream 2 supply	**5.0** bar	3.7 – 5.6
TIR 1.1.1	Air after MFCs	**21.4** °C	16.9 – 32.3
PIR 1.1.2	Air after MFCs	**250** mbar	131 – 290
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**581** m^3/h	549 – 593
TIR 2.04	TWV → APH inlet	**22.4** °C	18.7 – 31.4
Reformer air (POx)			
FIRC 1.2	POx air	**312** ln/min	240 – 491
PIR 1.2.1	POx air supply	**5.3** bar	4.3 – 6.1
TIR 1.2.1	POx air (FPM)	**20.0** °C	15.8 – 29.0
TIR 1.2.2	POx air FPM → PGM	**17.8** °C	15.3 – 20.6
PIR 1.2.2	POx air after MFC	**80.9** mbar	60.3 – 140
TIR 3.04	POx air PGM inlet	**19.6** °C	15.7 – 23.4
TIR 3.05	Reformer inlet POx air	**32.0** °C	26.9 – 38.0
Nitrogen			
FIRC 1.3	Nitrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.3.1	N_2 supply	**5.5** bar	5.5 – 5.6
TIR 1.3	Purge gas	**13.4** °C	10.1 – 16.8
PIR 1.3.2	Purge gas	**1.8** bar?	1.6 – 2.0
Hydrogen			
FIRC 1.5	Hydrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.5	H_2 supply	**5.4** bar	5.3 – 5.5

Natural gas			
FIRC 1.4	Natural gas	**137** ln/min	100.0 – 150
PIR 1.4.1	After booster	**13.4** bar	13.3 – 13.4
TIR 1.4.1	Fuel (FPM)	**15.5** °C	12.5 – 18.0
PIR 1.4.2	Fuel after MFC	**8.9** bar	6.4 – 10.0
TIR 1.4.2	Fuel FPM → PGM	**17.4** °C	15.6 – 19.8
TIR 3.01	Fuel PGM inlet	**16.4** °C	13.7 – 20.1
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**22.6** °C	18.8 – 31.1
TIR 2.02	APH1+2 outlet air	**585** °C	550 – 637
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**658** °C	614 – 684
PIR 7.1.1	APH inlet exhaust 1	**8.0** mbar	5.2 – 9.5
TIR 7.3.1	Flue gas out 1	**97.8** °C	89.3 – 156
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**22.3** °C	18.5 – 30.8
TIR 2.03	APH outlet riser → EAPH	**585** °C	550 – 634
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**664** °C	623 – 689
PIR 7.1.2	APH inlet exhaust 2	**8.0** mbar	5.1 – 9.5
TIR 7.3.2	Flue gas out 2	**94.9** °C	87.1 – 155
Air heater EAPH			
TIR 2.05	EAPH inlet	**582** °C	550 – 594
PIR 2.01	EAPH inlet	**122** mbar	67.5 – 142
EAPH power	Heater power (3-phase)	**8.6** kW	3.3 – 18.6
TIRZ 8.1.1	EAPH safety (trip 1050)	**592** °C	553 – 613
TIRC 2.06	EAPH outlet = stack air in	**600** °C	580 – 610
PIR 2.02	Stack module air inlet	**21.8** mbar	12.8 – 25.8

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**22.5** °C	17.9 – 25.4
PIR 3.01	Ejector main stream	**8.8** bar	6.3 – 10.0
TIR 6.01	Recycle gas inlet	**609** °C	567 – 636
PIR 6.01	Recycle gas inlet	**17.0** mbar	11.0 – 20.0
TIRC 3.03	Ejector outlet	**444** °C	390 – 487
PIR 3.02	Ejector outlet	**31.0** mbar	26.4 – 36.9
Reformer			
TIR 3.06	Reformer inlet mixed	**416** °C	322 – 466
TIRC 3.07	Catalyst inlet	**712** °C	681 – 809
TIRC 3.15	Catalyst outlet	**614** °C	592 – 644
TIRC 3.17	Reformer outlet	**619** °C	592 – 638
Heater ERH			
TIRC 5.17	ERH control	**702** °C	628 – 810
TIRZ 8.2.5	ERH safety (trip 860)	**712** °C	630 – 827
IIR 5.1.3	ERH current	**5.1** A	3.0 – 5.2
Heater EAGH			
TIRC 3.1.1	EAGH control	**610** °C	590 – 641
TIRZ 8.2.1	EAGH safety (trip 950)	**613** °C	591 – 643
IIR 3.1.1	EAGH current	**4.8** A	0.10 – 4.9
TIR 3.18	Stack module anode inlet	**616** °C	594 – 637
PIR 3.03	Stack module anode inlet	**22.2** mbar	17.1 – 26.6
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**625** °C	600 – 659
TIRZ 8.1.3	Stack safety (trip 720)	**642** °C	610 – 678
TIR 3.19.1	Manifold 1 inlet	**601** °C	582 – 619

TIR 3.19.2	Manifold 2 inlet	**597** °C	579 – 615
TIRC 5.1.1	Cathode outlet 1	**624** °C	597 – 657
TIRC 5.1.2	Cathode outlet 2	**628** °C	601 – 662
PIR 5.1.1	Cathode outlet 1	**19.2** mbar	11.0 – 22.7
TIRC 5.2.1	Anode outlet 1	**648** °C	617 – 691
TIRC 5.2.2	Anode outlet 2	**643** °C	600 – 666
PIR 5.2.1	Anode outlet 1	**17.1** mbar	11.2 – 20.4
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**6.5** A	2.0 – 14.0
EIRC 4.2.2	String 2 current	**6.3** A	1.9 – 13.9
EIRC 4.3.2	String 3 current	**6.5** A	2.1 – 14.1
EIRC 4.4.2	String 4 current	**5.5** A	1.5 – 9.2
EIRC 4.5.2	String 5 current	**6.3** A	1.9 – 13.7
EIRC 4.1.1	String 1 load voltage	**455** V	411 – 487
EIRC 4.2.1	String 2 load voltage	**460** V	414 – 491
EIRC 4.3.1	String 3 load voltage	**446** V	395 – 486
EIRC 4.4.1	String 4 load voltage	**455** V	436 – 480
EIRC 4.5.1	String 5 load voltage	**470** V	427 – 499
Σ stacks M1	String 1 stack-sum voltage	**430** V	390 – 458
Σ stacks M2	String 2 stack-sum voltage	**434** V	395 – 461
Σ stacks M3	String 3 stack-sum voltage	**422** V	377 – 459
Σ stacks M4	String 4 stack-sum voltage	**428** V	411 – 452
Σ stacks M5	String 5 stack-sum voltage	**444** V	406 – 469
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**512** °C	437 – 563
TIRZ 8.2.4	EAOH3 safety (trip 950)	**473** °C	395 – 527

IIR 6.1.1	EAOH3 current	**4.2** A	0.09 – 10.0
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**660** °C	618 – 709
TIRZ 8.2.2	EAOH1 safety (trip 950)	**660** °C	619 – 707
IIR 5.1.1	EAOH1 current	**3.0** A	0.09 – 5.2
TIRC 5.5.1	TOX inlet anode 1	**634** °C	595 – 696
PIR 5.4.1	TOX inlet anode 1	**16.6** mbar	10.7 – 19.8
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**658** °C	615 – 698
TIRZ 8.2.3	EAOH2 safety (trip 950)	**657** °C	616 – 695
IIR 5.1.2	EAOH2 current	**2.9** A	0.10 – 5.1
TIRC 5.5.2	TOX inlet anode 2	**621** °C	589 – 703
PIR 5.4.2	TOX inlet anode 2	**16.7** mbar	10.8 – 20.0
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**741** °C	678 – 790
TIRZ 8.1.9	Burner safety (trip 930)	**707** °C	643 – 755
TIR 5.4.1	TOX inlet cathode 1	**622** °C	597 – 655
PIR 5.3.1	TOX inlet cathode 1	**18.1** mbar	10.5 – 21.5
TIR 5.6.1	Cathode bypass 1	**616** °C	582 – 645
TIR 7.1.1	TOX outlet 1	**667** °C	616 – 701
TIR 7.1.2	TOX outlet 2	**694** °C	642 – 727
Exhaust			
TIR 7.04	APH exhaust outlet	**113** °C	106 – 188
PIR 7.02	APH exhaust outlet	**0.81** mbar	0.50 – 0.90
TIR 7.05	Exhaust (roof)	**112** °C	105 – 183
PIR 7.03	Exhaust	**0.48** mbar	0.32 – 0.55

TIRZ 8.1.10	Exhaust safety (trip 300)	**114** °C	107 – 183

If you remember one thing: Part load is the only stage with current; the stack heats itself and no run lasted long enough to settle.

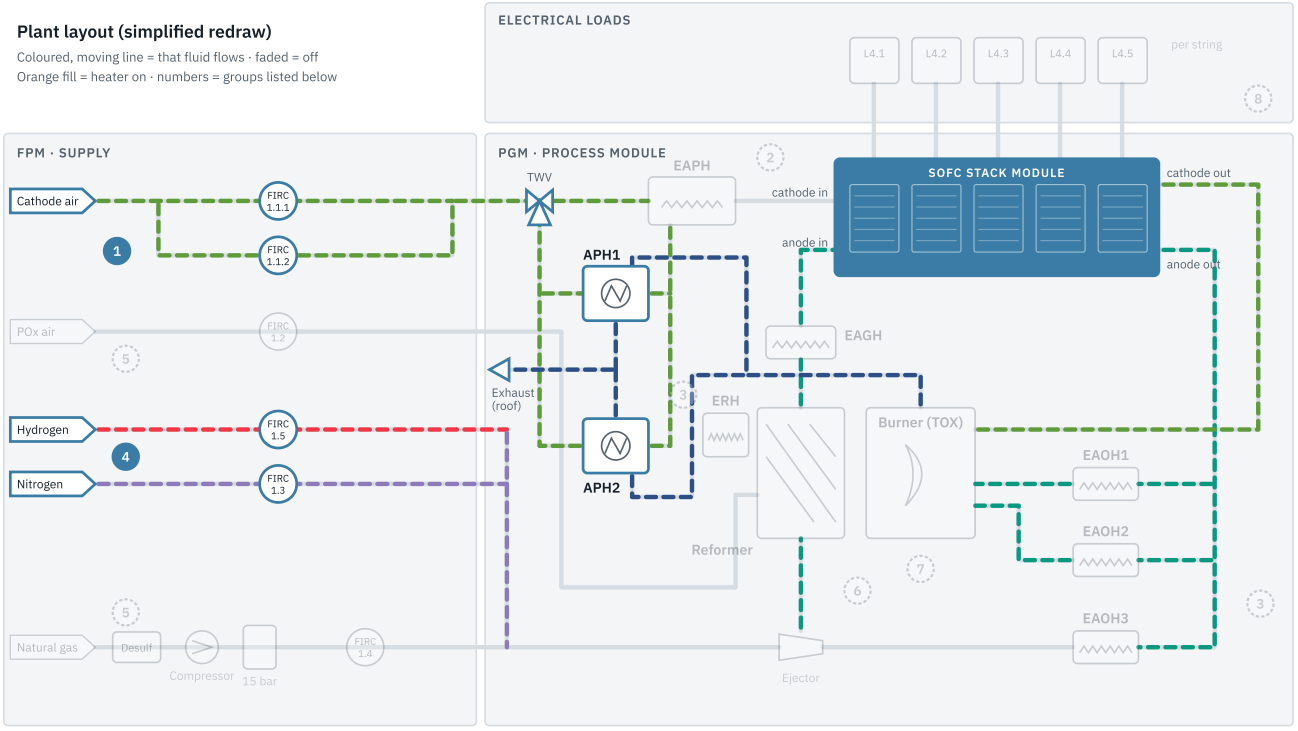
STAGE 6 OF 8 · SHUTTING DOWN, AIR STILL BLOWING

6 Air-flow cooling

> Gas and heaters are off. Air keeps blowing to carry the heat out at a safe speed. Forming gas stays on while the stack is above 210 °C.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
28	254.9 h	12 min	106.0 h

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000+

2 Electric air heater EAPH
Off

3 Gas-line heaters ERH · EAGH · EAOH 1–3
Off

4 Forming gas N_2 + H_2
On while the stack is above 210 °C

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

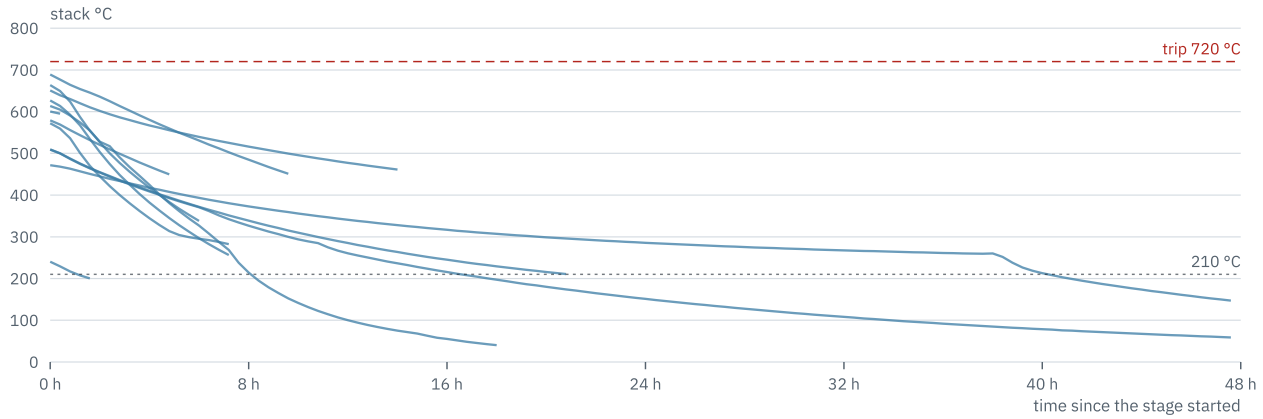
· Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000+ | ~~Electric air heater EAPH~~ | Recuperators APH (passive) | ~~Gas-line heaters ERH, EAGH, EAOH 1–2~~ | ~~Recycle heater EAOH 3~~ | Forming gas N_2 1200 + H_2 60 · > 210 °C (partly) | ~~Natural gas + reformer air~~ | ~~Reformer + gas recycle hot~~ | ~~After-burner burning~~ | ~~Stack current to loads~~ | Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

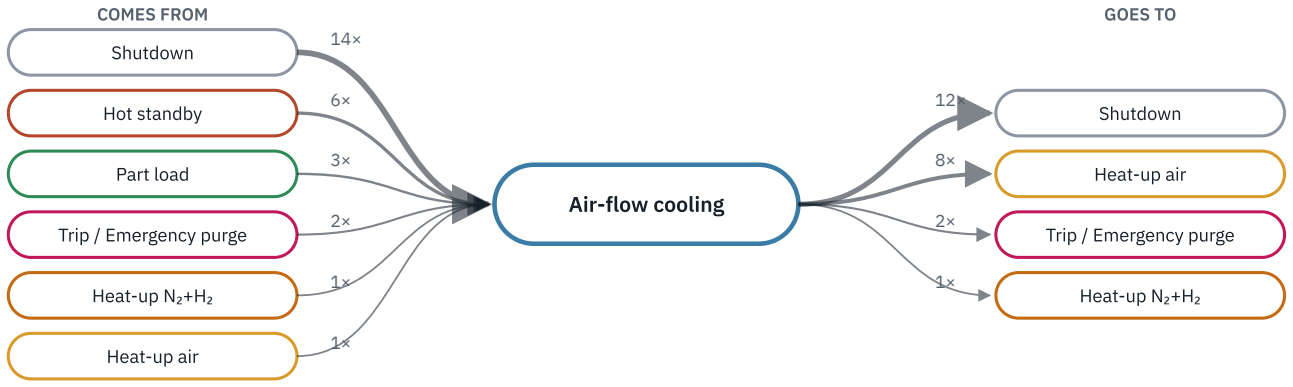


Every visit of this stage drawn from its start (one line per visit, 28 visits; shown up to 48.0 h). Cooling speed depends on the air: about −19 °C/h at 6000 ln/min and −45 °C/h at 8000–10000 ln/min.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**242** °C	42.6 – 514
Air into stack (TIRC 2.06)	**214** °C	39.6 – 448
Cathode air, both streams	**6,000** ln/min	1,261 – 8,998
Nitrogen N_2 (FIRC 1.3)	**1,200** ln/min	0.0 – 1,200
Hydrogen H_2 (FIRC 1.5)	**60.0** ln/min	0.0 – 60.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**437** V	14.8 – 479
Air heater EAPH power	**0.0** kW	0.0 – 0.10
Gas-line heater ERH current	**0.0** A	0.0 – 0.0
Recycle heater EAOH 3 current	**0.0** A	0.0 – 0.0
After-burner (TIRC 5.09)	**248** °C	43.6 – 484
Reformer catalyst inlet (TIRC 3.07)	**71.2** °C	35.1 – 163
Exhaust safety (TIRZ 8.1.10)	**61.5** °C	22.6 – 142

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. 255 h in total. Next: Shutdown (12 of 23) or a restart into Heat-up air (8).

Air-flow cooling → Heat-up air | Hot standby → Air-flow cooling | Air-flow cooling → Shutdown

Shutdown → Air-flow cooling | Part load → Air-flow cooling | Air-flow cooling → Trip / Emergency purge | Trip / Emergency purge → Air-flow cooling

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,000** ln/min	630 – 4,498
FIRC 1.1.2	Cathode air stream 2	**3,000** ln/min	630 – 4,498
PIR 1.1.1	Air stream 1 supply	**5.1** bar	5.0 – 5.6
PIR 1.1.3	Air stream 2 supply	**5.0** bar	4.8 – 5.5
TIR 1.1.1	Air after MFCs	**19.7** °C	17.6 – 20.3
PIR 1.1.2	Air after MFCs	**83.5** mbar	9.9 – 183
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**404** m³/h	31.4 – 527
TIR 2.04	TWV → APH inlet	**18.9** °C	17.5 – 20.1
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.4** bar	5.3 – 5.9
TIR 1.2.1	POx air (FPM)	**16.6** °C	13.4 – 18.3
TIR 1.2.2	POx air FPM → PGM	**4.5** °C	0.40 – 8.7
PIR 1.2.2	POx air after MFC	**57.0** mbar	1.9 – 67.1
TIR 3.04	POx air PGM inlet	**6.6** °C	3.1 – 12.1
TIR 3.05	Reformer inlet POx air	**15.5** °C	6.3 – 70.9
Nitrogen			
FIRC 1.3	Nitrogen	**1,200** ln/min	0.0 – 1,200
PIR 1.3.1	N_2 supply	**4.7** bar	4.7 – 5.7
TIR 1.3	Purge gas	**15.0** °C	2.1 – 17.4
PIR 1.3.2	Purge gas	**0.60** bar?	0.01 – 24.3
Hydrogen			
FIRC 1.5	Hydrogen	**60.0** ln/min	0.0 – 60.0
PIR 1.5	H_2 supply	**5.2** bar	4.8 – 5.4

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**12.1** bar	0.02 – 13.6
TIR 1.4.1	Fuel (FPM)	**16.9** °C	13.3 – 18.4
PIR 1.4.2	Fuel after MFC	**0.01** bar	0.00 – 0.02
TIR 1.4.2	Fuel FPM → PGM	**4.9** °C	0.80 – 9.1
TIR 3.01	Fuel PGM inlet	**11.4** °C	4.5 – 26.4
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**18.2** °C	16.5 – 19.9
TIR 2.02	APH1+2 outlet air	**218** °C	41.4 – 463
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**226** °C	43.9 – 476
PIR 7.1.1	APH inlet exhaust 1	**1.9** mbar	0.72 – 5.9
TIR 7.3.1	Flue gas out 1	**49.7** °C	19.7 – 132
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**17.9** °C	16.3 – 19.6
TIR 2.03	APH outlet riser → EAPH	**218** °C	40.8 – 463
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**230** °C	43.3 – 479
PIR 7.1.2	APH inlet exhaust 2	**5.3** mbar	4.3 – 7.9
TIR 7.3.2	Flue gas out 2	**48.4** °C	19.6 – 135
Air heater EAPH			
TIR 2.05	EAPH inlet	**217** °C	35.7 – 462
PIR 2.01	EAPH inlet	**26.2** mbar	5.7 – 66.4
EAPH power	Heater power (3-phase)	**0.0** kW	0.0 – 0.09
TIRZ 8.1.1	EAPH safety (trip 1050)	**214** °C	40.4 – 443
TIRC 2.06	EAPH outlet = stack air in	**214** °C	39.6 – 448
PIR 2.02	Stack module air inlet	**4.6** mbar	2.1 – 14.7

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**25.4** °C	16.9 – 39.1
PIR 3.01	Ejector main stream	**0.01** bar	0.00 – 0.02
TIR 6.01	Recycle gas inlet	**52.1** °C	12.3 – 125
PIR 6.01	Recycle gas inlet	**6.4** mbar	2.1 – 15.6
TIRC 3.03	Ejector outlet	**20.3** °C	7.6 – 71.8
PIR 3.02	Ejector outlet	**8.5** mbar	1.8 – 19.3
Reformer			
TIR 3.06	Reformer inlet mixed	**25.4** °C	10.2 – 95.9
TIRC 3.07	Catalyst inlet	**71.2** °C	35.1 – 163
TIRC 3.15	Catalyst outlet	**183** °C	42.8 – 448
TIRC 3.17	Reformer outlet	**209** °C	51.1 – 477
Heater ERH			
TIRC 5.17	ERH control	**223** °C	44.1 – 506
TIRZ 8.2.5	ERH safety (trip 860)	**226** °C	46.7 – 507
IIR 5.1.3	ERH current	**0.00** A	0.0 – 0.00
Heater EAGH			
TIRC 3.1.1	EAGH control	**202** °C	55.3 – 504
TIRZ 8.2.1	EAGH safety (trip 950)	**205** °C	54.4 – 501
IIR 3.1.1	EAGH current	**0.00** A	0.0 – 0.00
TIR 3.18	Stack module anode inlet	**200** °C	48.0 – 491
PIR 3.03	Stack module anode inlet	**7.5** mbar	2.3 – 17.2
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**242** °C	42.6 – 514
TIRZ 8.1.3	Stack safety (trip 720)	**249** °C	44.7 – 525
TIR 3.19.1	Manifold 1 inlet	**201** °C	54.7 – 489

TIR 3.19.2	Manifold 2 inlet	**201** °C	53.4 – 491
TIRC 5.1.1	Cathode outlet 1	**244** °C	43.1 – 514
TIRC 5.1.2	Cathode outlet 2	**243** °C	43.4 – 517
PIR 5.1.1	Cathode outlet 1	**3.8** mbar	1.9 – 13.1
TIRC 5.2.1	Anode outlet 1	**256** °C	47.3 – 549
TIRC 5.2.2	Anode outlet 2	**250** °C	45.9 – 532
PIR 5.2.1	Anode outlet 1	**6.1** mbar	1.7 – 15.3
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.0
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.12** V	0.08 – 0.27
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**437** V	14.8 – 479
Σ stacks M2	String 2 stack-sum voltage	**446** V	9.3 – 486
Σ stacks M3	String 3 stack-sum voltage	**430** V	11.1 – 478
Σ stacks M4	String 4 stack-sum voltage	**438** V	10.4 – 473
Σ stacks M5	String 5 stack-sum voltage	**470** V	5.1 – 499
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**53.3** °C	15.7 – 114
TIRZ 8.2.4	EAOH3 safety (trip 950)	**56.2** °C	17.2 – 120

IIR 6.1.1	EAOH3 current	**0.00** A	0.0 – 0.01
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**233** °C	61.2 – 495
TIRZ 8.2.2	EAOH1 safety (trip 950)	**232** °C	63.6 – 505
IIR 5.1.1	EAOH1 current	**0.00** A	0.0 – 0.00
TIRC 5.5.1	TOX inlet anode 1	**169** °C	55.8 – 347
PIR 5.4.1	TOX inlet anode 1	**5.5** mbar	2.0 – 14.6
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**230** °C	62.4 – 514
TIRZ 8.2.3	EAOH2 safety (trip 950)	**235** °C	64.3 – 526
IIR 5.1.2	EAOH2 current	**0.00** A	0.0 – 0.00
TIRC 5.5.2	TOX inlet anode 2	**173** °C	44.6 – 359
PIR 5.4.2	TOX inlet anode 2	**5.6** mbar	2.2 – 14.8
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**248** °C	43.6 – 484
TIRZ 8.1.9	Burner safety (trip 930)	**241** °C	45.2 – 478
TIR 5.4.1	TOX inlet cathode 1	**245** °C	43.1 – 514
PIR 5.3.1	TOX inlet cathode 1	**3.7** mbar	1.9 – 12.6
TIR 5.6.1	Cathode bypass 1	**217** °C	43.8 – 475
TIR 7.1.1	TOX outlet 1	**234** °C	43.6 – 493
TIR 7.1.2	TOX outlet 2	**232** °C	43.1 – 480
Exhaust			
TIR 7.04	APH exhaust outlet	**60.7** °C	20.8 – 143
PIR 7.02	APH exhaust outlet	**0.35** mbar	0.10 – 0.65
TIR 7.05	Exhaust (roof)	**59.8** °C	20.5 – 142
PIR 7.03	Exhaust	**0.17** mbar	0.02 – 0.37

TIRZ 8.1.10	Exhaust safety (trip 300)	**61.5** °C	22.6 – 142

If you remember one thing: Air-flow cooling is the heaters off and the air carrying the heat out; more air means faster cooling.

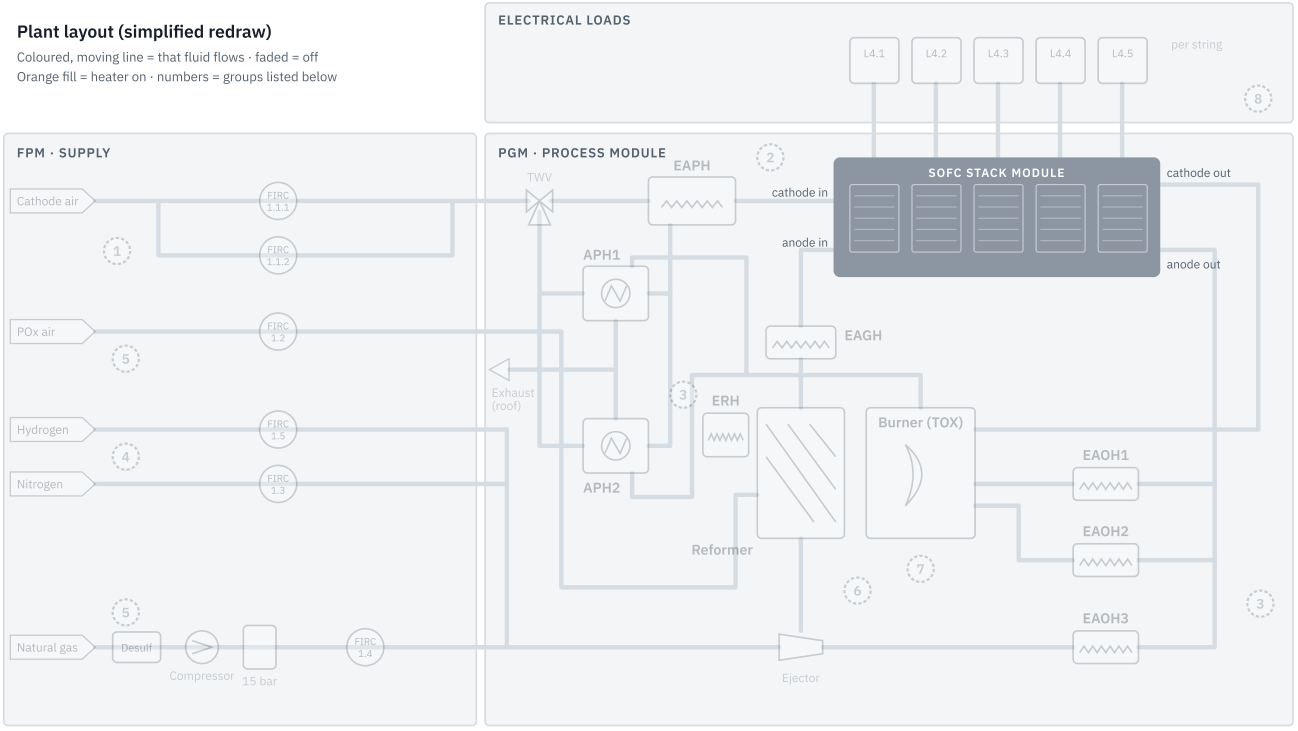
STAGE 7 OF 8 · EVERYTHING OFF EXCEPT THE CONTROLS

7 Shutdown

> Air, gases, heaters and current are all off. Covers a cold plant, a warm plant left to cool with no air, and the first minutes after an emergency stop.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
33	173.7 h	3.0 h	16.4 h

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1. Cathode air + recuperators (APH)
   Off
2. Electric air heater EAPH
   Off
3. Gas-line heaters ERH · EAGH · EAOH 1–3
   Off
4. Forming gas N_2 + H_2
   Off

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

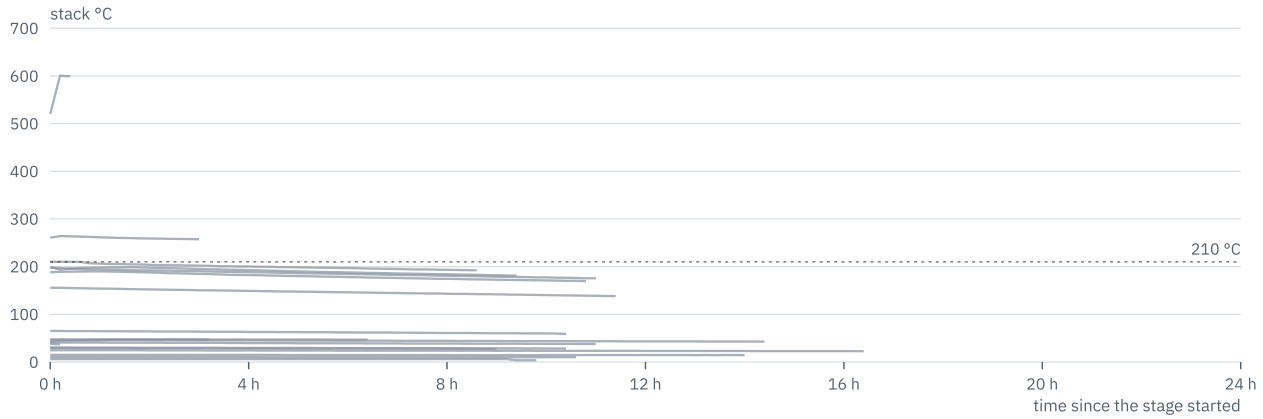
· Controls (PLC, safety)
Always on · not drawn

What is switched on

~~Cathode air (2 streams)~~ ~~Electric air heater EAPH~~ ~~Recuperators APH (passive)~~ ~~Gas-line heaters ERH, EAGH, EAOH 1–2~~ ~~Recycle heater EAOH 3~~ ~~Forming gas N_2 1200 + H_2 60~~ ~~Natural gas + reformer air~~ ~~Reformer + gas recycle hot~~ ~~After-burner burning~~ ~~Stack current to loads~~ Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

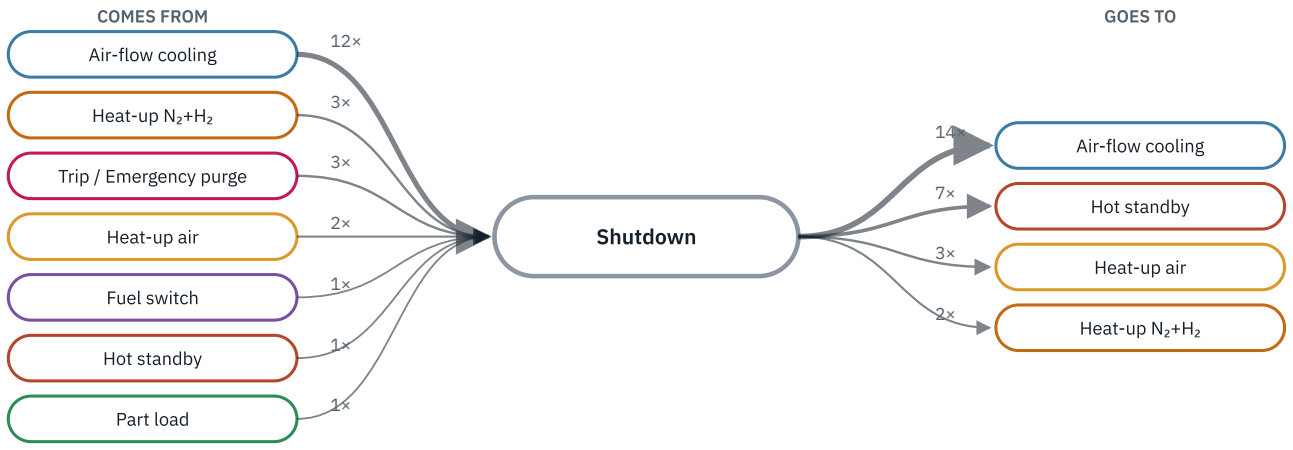


Every visit of this stage drawn from its start (one line per visit, 33 visits; shown up to 24.0 h). Very slow drift towards room temperature. The hot box holds its heat for days without air.

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**43.7** °C	10.2 – 193
Air into stack (TIRC 2.06)	**42.4** °C	9.4 – 192
Cathode air, both streams	**0.0** ln/min	0.0 – 0.0
Nitrogen N_2 (FIRC 1.3)	**0.0** ln/min	0.0 – 0.0
Hydrogen H_2 (FIRC 1.5)	**0.0** ln/min	0.0 – 0.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**0.80** V	0.40 – 9.9
Air heater EAPH power	**0.0** kW	0.0 – 0.10
Gas-line heater ERH current	**0.0** A	0.0 – 0.0
Recycle heater EAOH 3 current	**0.0** A	0.0 – 0.0
After-burner (TIRC 5.09)	**40.5** °C	10.3 – 176
Reformer catalyst inlet (TIRC 3.07)	**39.6** °C	9.5 – 175
Exhaust safety (TIRZ 8.1.10)	**10.4** °C	7.6 – 20.3

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. 174 h in total. Next: Air-flow cooling (14), Hot standby (7) or Heat-up air (4).

Shutdown → Heat-up air | Shutdown → Heat-up N_2+H_2 | Shutdown → Hot standby | Air-flow cooling → Shutdown | Shutdown → Air-flow cooling | Heat-up air → Shutdown | Heat-up N_2+H_2 → Shutdown | Trip / Emergency purge → Shutdown

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**0.0** ln/min	0.0 – 0.0
FIRC 1.1.2	Cathode air stream 2	**0.0** ln/min	0.0 – 0.0
PIR 1.1.1	Air stream 1 supply	**5.4** bar	5.2 – 6.3
PIR 1.1.3	Air stream 2 supply	**5.4** bar	5.2 – 6.3
TIR 1.1.1	Air after MFCs	**16.8** °C	9.8 – 18.4
PIR 1.1.2	Air after MFCs	**0.20** mbar	0.14 – 0.34
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**54.3** m³/h	39.4 – 132
TIR 2.04	TWV → APH inlet	**6.1** °C	2.3 – 10.9
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.4** bar	5.3 – 6.2
TIR 1.2.1	POx air (FPM)	**16.1** °C	9.1 – 17.7
TIR 1.2.2	POx air FPM → PGM	**4.3** °C	0.79 – 8.4
PIR 1.2.2	POx air after MFC	**0.01** mbar	0.0 – 0.12
TIR 3.04	POx air PGM inlet	**5.6** °C	0.99 – 10.1
TIR 3.05	Reformer inlet POx air	**26.9** °C	9.8 – 107
Nitrogen			
FIRC 1.3	Nitrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.3.1	N_2 supply	**5.6** bar	5.4 – 5.8
TIR 1.3	Purge gas	**15.4** °C	7.9 – 17.2
PIR 1.3.2	Purge gas	**0.33** bar?	0.01 – 0.41
Hydrogen			
FIRC 1.5	Hydrogen	**0.0** ln/min	0.0 – 0.0
PIR 1.5	H_2 supply	**5.3** bar	5.0 – 5.5

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**0.02** bar	0.01 – 13.0
TIR 1.4.1	Fuel (FPM)	**16.3** °C	8.9 – 18.0
PIR 1.4.2	Fuel after MFC	**0.00** bar	0.0 – 0.00
TIR 1.4.2	Fuel FPM → PGM	**4.6** °C	0.40 – 8.9
TIR 3.01	Fuel PGM inlet	**5.9** °C	1.2 – 10.3
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**15.2** °C	8.3 – 31.1
TIR 2.02	APH1+2 outlet air	**43.8** °C	10.1 – 158
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**32.5** °C	9.8 – 135
PIR 7.1.1	APH inlet exhaust 1	**0.08** mbar	0.03 – 0.14
TIR 7.3.1	Flue gas out 1	**21.3** °C	9.3 – 39.7
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**14.7** °C	8.1 – 31.1
TIR 2.03	APH outlet riser → EAPH	**44.0** °C	9.7 – 133
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**31.1** °C	9.6 – 132
PIR 7.1.2	APH inlet exhaust 2	**0.01** mbar	0.0 – 0.07
TIR 7.3.2	Flue gas out 2	**20.8** °C	9.1 – 39.4
Air heater EAPH			
TIR 2.05	EAPH inlet	**43.2** °C	10.2 – 139
PIR 2.01	EAPH inlet	**0.64** mbar	0.55 – 0.69
EAPH power	Heater power (3-phase)	**0.05** kW	0.0 – 0.10
TIRZ 8.1.1	EAPH safety (trip 1050)	**40.1** °C	11.0 – 181
TIRC 2.06	EAPH outlet = stack air in	**42.4** °C	9.4 – 192
PIR 2.02	Stack module air inlet	**0.33** mbar	0.28 – 0.40

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**11.9** °C	6.1 – 41.1
PIR 3.01	Ejector main stream	**0.0** bar	0.0 – 0.00
TIR 6.01	Recycle gas inlet	**56.5** °C	9.8 – 188
PIR 6.01	Recycle gas inlet	**0.21** mbar	0.17 – 0.25
TIRC 3.03	Ejector outlet	**34.9** °C	9.8 – 149
PIR 3.02	Ejector outlet	**0.0** mbar	0.0 – 0.01
Reformer			
TIR 3.06	Reformer inlet mixed	**37.0** °C	9.7 – 153
TIRC 3.07	Catalyst inlet	**39.6** °C	9.5 – 175
TIRC 3.15	Catalyst outlet	**40.2** °C	9.6 – 181
TIRC 3.17	Reformer outlet	**41.1** °C	10.1 – 194
Heater ERH			
TIRC 5.17	ERH control	**41.8** °C	9.4 – 195
TIRZ 8.2.5	ERH safety (trip 860)	**44.2** °C	11.4 – 196
IIR 5.1.3	ERH current	**0.00** A	0.0 – 0.00
Heater EAGH			
TIRC 3.1.1	EAGH control	**41.6** °C	10.4 – 198
TIRZ 8.2.1	EAGH safety (trip 950)	**42.1** °C	10.6 – 193
IIR 3.1.1	EAGH current	**0.00** A	0.0 – 0.00
TIR 3.18	Stack module anode inlet	**40.5** °C	10.1 – 169
PIR 3.03	Stack module anode inlet	**0.37** mbar	0.31 – 0.45
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**43.7** °C	10.2 – 193
TIRZ 8.1.3	Stack safety (trip 720)	**45.7** °C	11.7 – 196
TIR 3.19.1	Manifold 1 inlet	**47.3** °C	10.1 – 164

TIR 3.19.2	Manifold 2 inlet	**47.3** °C	10.8 – 152
TIRC 5.1.1	Cathode outlet 1	**44.1** °C	10.4 – 179
TIRC 5.1.2	Cathode outlet 2	**42.0** °C	11.0 – 176
PIR 5.1.1	Cathode outlet 1	**0.15** mbar	0.08 – 0.21
TIRC 5.2.1	Anode outlet 1	**53.8** °C	10.0 – 188
TIRC 5.2.2	Anode outlet 2	**46.8** °C	9.0 – 192
PIR 5.2.1	Anode outlet 1	**0.0** mbar	0.0 – 0.01
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.00
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.14** V	0.0 – 0.30
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**0.75** V	0.38 – 9.9
Σ stacks M2	String 2 stack-sum voltage	**0.68** V	0.32 – 11.2
Σ stacks M3	String 3 stack-sum voltage	**0.52** V	0.21 – 11.1
Σ stacks M4	String 4 stack-sum voltage	**0.57** V	0.28 – 12.3
Σ stacks M5	String 5 stack-sum voltage	**0.33** V	0.19 – 12.5
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**50.4** °C	10.6 – 195
TIRZ 8.2.4	EAOH3 safety (trip 950)	**53.3** °C	12.7 – 192

IIR 6.1.1	EAOH3 current	**0.00** A	0.0 – 0.01
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**59.6** °C	11.3 – 206
TIRZ 8.2.2	EAOH1 safety (trip 950)	**61.2** °C	13.2 – 206
IIR 5.1.1	EAOH1 current	**0.00** A	0.0 – 0.00
TIRC 5.5.1	TOX inlet anode 1	**41.8** °C	9.7 – 193
PIR 5.4.1	TOX inlet anode 1	**0.15** mbar	0.13 – 0.17
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**59.0** °C	11.2 – 215
TIRZ 8.2.3	EAOH2 safety (trip 950)	**60.8** °C	13.3 – 214
IIR 5.1.2	EAOH2 current	**0.00** A	0.0 – 0.00
TIRC 5.5.2	TOX inlet anode 2	**41.0** °C	9.3 – 176
PIR 5.4.2	TOX inlet anode 2	**0.30** mbar	0.27 – 0.35
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**40.5** °C	10.3 – 176
TIRZ 8.1.9	Burner safety (trip 930)	**43.3** °C	11.8 – 185
TIR 5.4.1	TOX inlet cathode 1	**42.0** °C	10.1 – 173
PIR 5.3.1	TOX inlet cathode 1	**0.29** mbar	0.24 – 0.33
TIR 5.6.1	Cathode bypass 1	**35.7** °C	10.2 – 149
TIR 7.1.1	TOX outlet 1	**40.4** °C	9.6 – 177
TIR 7.1.2	TOX outlet 2	**38.3** °C	9.7 – 170
Exhaust			
TIR 7.04	APH exhaust outlet	**21.6** °C	9.6 – 43.4
PIR 7.02	APH exhaust outlet	**0.15** mbar	0.13 – 0.17
TIR 7.05	Exhaust (roof)	**14.1** °C	8.2 – 29.7
PIR 7.03	Exhaust	**0.07** mbar	0.04 – 0.10

TIRZ 8.1.10	Exhaust safety (trip 300)	**10.4** °C	7.6 – 20.3

If you remember one thing: Shutdown is only the controls on; a warm stack loses about 2 °C per hour without air.

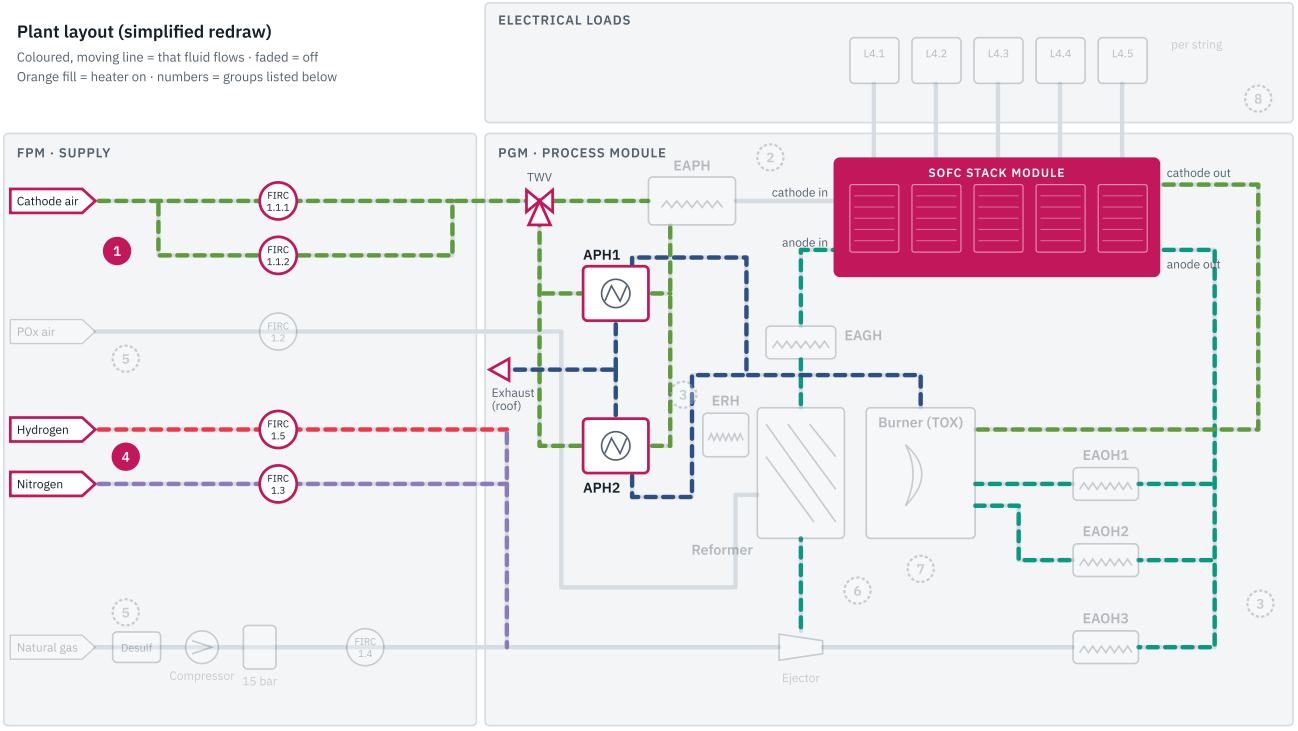
STAGE 8 OF 8 · SAFETY STOP

8 Trip / Emergency purge

A reading crossed a safety limit and the plant cut itself off. Only air, nitrogen and hydrogen keep flowing to flush the stack safely.

VISITS	TOTAL TIME	TYPICAL VISIT	LONGEST VISIT
6	1.3 h	15 min	17 min

Plant layout: which groups are working



The same simplified redraw of the plant as the digital twin app. Groups that work in this stage are outlined in the stage colour with their fluid lines moving; the rest are faded. On/off comes from the plant medians in this stage (the same table as the switch list below).

1 Cathode air + recuperators (APH)
On · 6000

2 Electric air heater EAPH
Off

3 Gas-line heaters ERH · EAGH · EAOH 1–3
Off

4 Forming gas N_2 + H_2
On

5 Natural gas + reformer air (POx)
Off

6 Reformer + gas recycle (ejector)
Off

7 After-burner (TOX) burning
Off

8 Stack current to the loads (power out)
Off

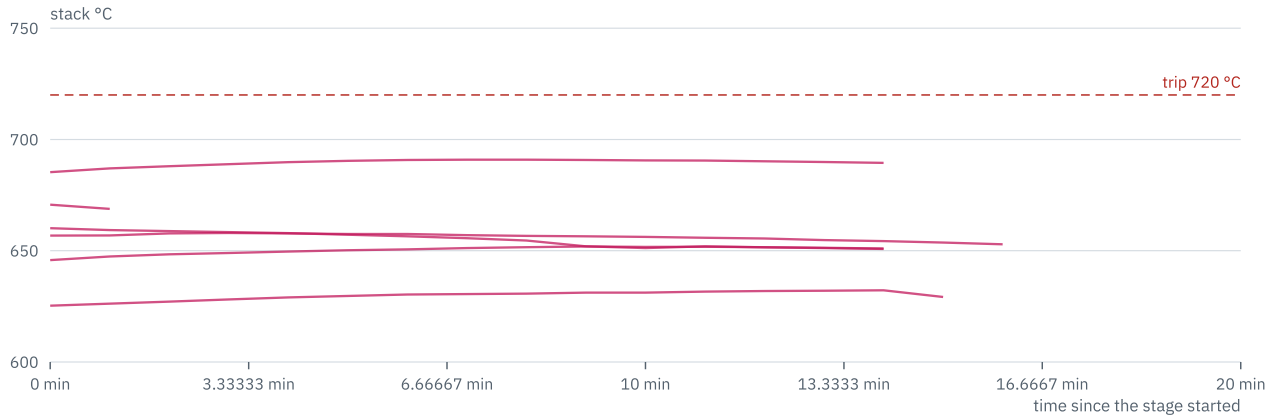
Controls (PLC, safety)
Always on · not drawn

What is switched on

Cathode air (2 streams) · 6000
~~Electric air heater EAPH~~
Recuperators APH (passive)
~~Gas-line heaters ERH, EAGH, EAOH 1–2~~
~~Recycle heater EAOH 3~~
Forming gas N_2 1200 + H_2 60
~~Natural gas + reformer air~~
~~Reformer + gas recycle hot~~
~~After-burner burning~~
~~Stack current to loads~~
Controls

Struck-through = off. Values are plant medians; heaters below full power count as on.

How the stack temperature moves in this stage

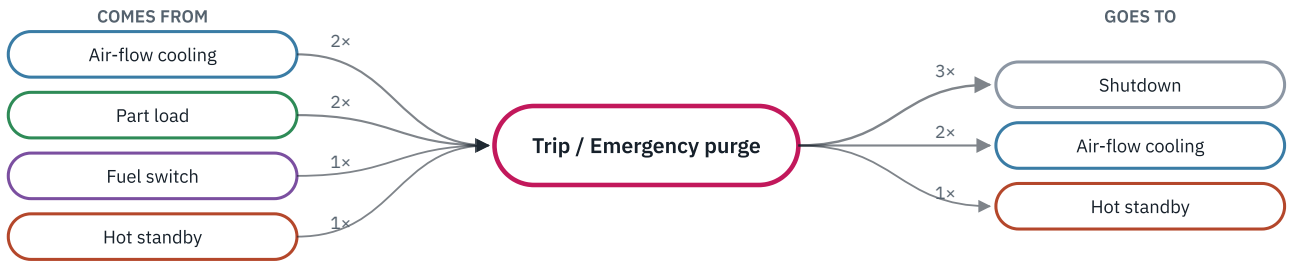


Every visit of this stage drawn from its start (one line per visit, 6 visits; shown up to 20 min). Most trips came from the exhaust safety reading TIRZ 8.1.10 going just over its limit seconds before the cut (6 of 9 trips).

Key readings in this stage

READING	MEDIAN	TYPICAL RANGE (10–90 %)
Stack temperature	**654** °C	630 – 690
Air into stack (TIRC 2.06)	**598** °C	569 – 634
Cathode air, both streams	**6,000** ln/min	5,950 – 6,001
Nitrogen N_2 (FIRC 1.3)	**1,200** ln/min	432 – 1,201
Hydrogen H_2 (FIRC 1.5)	**60.0** ln/min	21.2 – 60.0
Natural gas (FIRC 1.4)	**0.0** ln/min	0.0 – 0.0
Reformer air (FIRC 1.2)	**0.0** ln/min	0.0 – 0.0
String 1 current	**0.0** A	0.0 – 0.0
String 1 stack voltage	**412** V	384 – 418
Air heater EAPH power	**0.0** kW	0.0 – 3.2
Gas-line heater ERH current	**0.0** A	0.0 – 0.70
Recycle heater EAOH 3 current	**0.0** A	0.0 – 1.5
After-burner (TIRC 5.09)	**621** °C	596 – 661
Reformer catalyst inlet (TIRC 3.07)	**242** °C	218 – 411
Exhaust safety (TIRZ 8.1.10)	**176** °C	161 – 198

Where it comes from and where it goes



Counted between two stretches of logged data. Click a stage to open its page. 6 purges, 1.3 h in total. Next: Shutdown (3), Air-flow cooling (2) or Hot standby (1).

Part load → Trip / Emergency purge

Air-flow cooling → Trip / Emergency purge

Trip / Emergency purge → Air-flow cooling

Trip / Emergency purge → Shutdown

The links above open the matching change on the two change pages.

All 111 twin sensors in this stage

The same sensors the digital twin shows, with the plant's median and typical range in this stage.

TAG	READING	MEDIAN	TYPICAL RANGE (10–90 %)
Cathode air			
FIRC 1.1.1	Cathode air stream 1	**3,000** ln/min	2,975 – 3,001
FIRC 1.1.2	Cathode air stream 2	**3,000** ln/min	2,975 – 3,001
PIR 1.1.1	Air stream 1 supply	**5.1** bar	5.0 – 5.9
PIR 1.1.3	Air stream 2 supply	**5.0** bar	4.9 – 5.9
TIR 1.1.1	Air after MFCs	**19.8** °C	16.8 – 23.8
PIR 1.1.2	Air after MFCs	**136** mbar	130 – 141
3-way valve			
FIRC 1.1.3	3-way valve (temperature-driven)	**607** m^3/h	588 – 628
TIR 2.04	TWV → APH inlet	**21.8** °C	19.1 – 24.9
Reformer air (POx)			
FIRC 1.2	POx air	**0.0** ln/min	0.0 – 0.0
PIR 1.2.1	POx air supply	**5.8** bar	5.6 – 6.0
TIR 1.2.1	POx air (FPM)	**19.4** °C	16.5 – 21.9
TIR 1.2.2	POx air FPM → PGM	**14.4** °C	11.7 – 18.5
PIR 1.2.2	POx air after MFC	**70.7** mbar	19.4 – 72.3
TIR 3.04	POx air PGM inlet	**15.1** °C	11.9 – 19.7
TIR 3.05	Reformer inlet POx air	**22.5** °C	17.8 – 32.2
Nitrogen			
FIRC 1.3	Nitrogen	**1,200** ln/min	432 – 1,201
PIR 1.3.1	N_2 supply	**4.6** bar	4.6 – 5.1
TIR 1.3	Purge gas	**11.0** °C	6.6 – 15.8
PIR 1.3.2	Purge gas	**24.5** bar?	5.2 – 24.8
Hydrogen			
FIRC 1.5	Hydrogen	**60.0** ln/min	21.2 – 60.0
PIR 1.5	H_2 supply	**4.7** bar	4.7 – 4.8

Natural gas			
FIRC 1.4	Natural gas	**0.0** ln/min	0.0 – 0.0
PIR 1.4.1	After booster	**13.7** bar	13.6 – 13.7
TIR 1.4.1	Fuel (FPM)	**17.4** °C	15.2 – 19.8
PIR 1.4.2	Fuel after MFC	**0.03** bar	0.02 – 0.03
TIR 1.4.2	Fuel FPM → PGM	**14.1** °C	11.6 – 18.1
TIR 3.01	Fuel PGM inlet	**22.7** °C	18.2 – 26.3
Recuperator APH 1			
TIR 2.1.1	APH1 inlet air	**22.1** °C	19.1 – 25.1
TIR 2.02	APH1+2 outlet air	**629** °C	603 – 648
TIR 7.2.1	Flue gas in (TOX outlet mixed 1)	**640** °C	615 – 672
PIR 7.1.1	APH inlet exhaust 1	**6.3** mbar	5.5 – 6.7
TIR 7.3.1	Flue gas out 1	**150** °C	136 – 170
Recuperator APH 2			
TIR 2.1.2	APH2 inlet air	**21.9** °C	18.8 – 24.9
TIR 2.03	APH outlet riser → EAPH	**626** °C	596 – 647
TIR 7.2.2	Flue gas in (TOX outlet mixed 2)	**641** °C	616 – 676
PIR 7.1.2	APH inlet exhaust 2	**6.2** mbar	5.5 – 6.6
TIR 7.3.2	Flue gas out 2	**148** °C	135 – 169
Air heater EAPH			
TIR 2.05	EAPH inlet	**614** °C	589 – 643
PIR 2.01	EAPH inlet	**70.9** mbar	64.3 – 75.0
EAPH power	Heater power (3-phase)	**0.01** kW	0.0 – 3.2
TIRZ 8.1.1	EAPH safety (trip 1050)	**591** °C	542 – 635
TIRC 2.06	EAPH outlet = stack air in	**598** °C	569 – 634
PIR 2.02	Stack module air inlet	**15.3** mbar	13.0 – 15.8

Ejector / gas recycle			
TIR 3.02	Ejector main stream	**26.9** °C	21.6 – 30.1
PIR 3.01	Ejector main stream	**0.03** bar	0.02 – 0.03
TIR 6.01	Recycle gas inlet	**230** °C	178 – 385
PIR 6.01	Recycle gas inlet	**16.9** mbar	11.1 – 17.5
TIRC 3.03	Ejector outlet	**49.4** °C	32.6 – 202
PIR 3.02	Ejector outlet	**22.0** mbar	12.3 – 22.9
Reformer			
TIR 3.06	Reformer inlet mixed	**51.3** °C	42.3 – 171
TIRC 3.07	Catalyst inlet	**242** °C	218 – 411
TIRC 3.15	Catalyst outlet	**629** °C	612 – 665
TIRC 3.17	Reformer outlet	**637** °C	616 – 667
Heater ERH			
TIRC 5.17	ERH control	**705** °C	640 – 761
TIRZ 8.2.5	ERH safety (trip 860)	**704** °C	643 – 760
IIR 5.1.3	ERH current	**0.00** A	0.00 – 0.73
Heater EAGH			
TIRC 3.1.1	EAGH control	**622** °C	601 – 650
TIRZ 8.2.1	EAGH safety (trip 950)	**625** °C	604 – 655
IIR 3.1.1	EAGH current	**0.00** A	0.00 – 1.1
TIR 3.18	Stack module anode inlet	**638** °C	607 – 666
PIR 3.03	Stack module anode inlet	**18.7** mbar	11.7 – 19.4
Stack			
TIR 4.6.x	Stack median of 36 TCs (model)	**654** °C	630 – 690
TIRZ 8.1.3	Stack safety (trip 720)	**667** °C	644 – 700
TIR 3.19.1	Manifold 1 inlet	**615** °C	596 – 637

TIR 3.19.2	Manifold 2 inlet	**611** °C	593 – 632
TIRC 5.1.1	Cathode outlet 1	**649** °C	627 – 682
TIRC 5.1.2	Cathode outlet 2	**651** °C	631 – 686
PIR 5.1.1	Cathode outlet 1	**13.4** mbar	11.3 – 14.0
TIRC 5.2.1	Anode outlet 1	**640** °C	618 – 661
TIRC 5.2.2	Anode outlet 2	**637** °C	614 – 670
PIR 5.2.1	Anode outlet 1	**16.6** mbar	10.8 – 17.2
Loads (current, voltage)			
EIRC 4.1.2	String 1 current	**0.0** A	0.0 – 0.0
EIRC 4.2.2	String 2 current	**0.0** A	0.0 – 0.0
EIRC 4.3.2	String 3 current	**0.0** A	0.0 – 0.0
EIRC 4.4.2	String 4 current	**0.0** A	0.0 – 0.0
EIRC 4.5.2	String 5 current	**0.0** A	0.0 – 0.0
EIRC 4.1.1	String 1 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.2.1	String 2 load voltage	**0.25** V	0.15 – 0.29
EIRC 4.3.1	String 3 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.4.1	String 4 load voltage	**0.0** V	0.0 – 0.0
EIRC 4.5.1	String 5 load voltage	**0.0** V	0.0 – 0.0
Σ stacks M1	String 1 stack-sum voltage	**412** V	385 – 418
Σ stacks M2	String 2 stack-sum voltage	**416** V	388 – 423
Σ stacks M3	String 3 stack-sum voltage	**417** V	385 – 421
Σ stacks M4	String 4 stack-sum voltage	**401** V	379 – 409
Σ stacks M5	String 5 stack-sum voltage	**451** V	429 – 454
Recycle heater EAOH 3			
TIRC 6.1.1	EAOH3 control	**322** °C	256 – 483
TIRZ 8.2.4	EAOH3 safety (trip 950)	**308** °C	246 – 458

IIR 6.1.1	EAOH3 current	**0.00** A	0.00 – 1.5
Heater EAOH 1			
TIRC 5.3.1	EAOH1 control	**641** °C	608 – 685
TIRZ 8.2.2	EAOH1 safety (trip 950)	**643** °C	611 – 685
IIR 5.1.1	EAOH1 current	**0.00** A	0.00 – 0.66
TIRC 5.5.1	TOX inlet anode 1	**529** °C	494 – 641
PIR 5.4.1	TOX inlet anode 1	**15.6** mbar	10.6 – 16.2
Heater EAOH 2			
TIRC 5.3.2	EAOH2 control	**640** °C	610 – 681
TIRZ 8.2.3	EAOH2 safety (trip 950)	**642** °C	614 – 681
IIR 5.1.2	EAOH2 current	**0.00** A	0.00 – 0.72
TIRC 5.5.2	TOX inlet anode 2	**539** °C	504 – 646
PIR 5.4.2	TOX inlet anode 2	**15.8** mbar	10.7 – 16.3
After-burner (TOX)			
TIRC 5.09	Burner (TOX) representative	**621** °C	596 – 661
TIRZ 8.1.9	Burner safety (trip 930)	**625** °C	600 – 668
TIR 5.4.1	TOX inlet cathode 1	**646** °C	624 – 677
PIR 5.3.1	TOX inlet cathode 1	**12.9** mbar	10.9 – 13.4
TIR 5.6.1	Cathode bypass 1	**621** °C	600 – 645
TIR 7.1.1	TOX outlet 1	**650** °C	629 – 683
TIR 7.1.2	TOX outlet 2	**641** °C	618 – 678
Exhaust			
TIR 7.04	APH exhaust outlet	**177** °C	162 – 203
PIR 7.02	APH exhaust outlet	**0.58** mbar	0.51 – 0.67
TIR 7.05	Exhaust (roof)	**175** °C	161 – 197
PIR 7.03	Exhaust	**0.38** mbar	0.34 – 0.44

TIRZ 8.1.10	Exhaust safety (trip 300)	**176** °C	161 – 198

If you remember one thing: The trip purge is a 15–17 minute safety flush with air, N_2 and H_2 only.

Starting up and loading

> On the way up each move **switches one supply on**: air, then forming gas, then natural gas, then current. Flows and the voltage jump within a minute; temperatures bend at once but need hours to arrive.

How to read each change

Before = median of the last 15 min of the old stage · Early in next stage = about 30 min in (or the end of the next stage, if it is shorter) · End of next stage = its last 15 min · Starts to move = minutes until the reading has made a tenth of its move. ▲ rises, ▼ falls. Only readings that really change are listed.

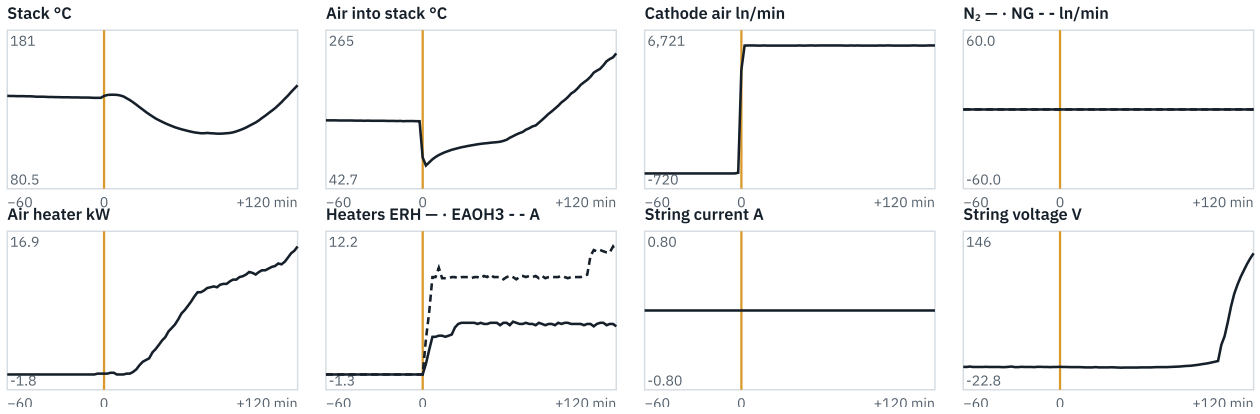
Shutdown → Heat-up air 3×
Air-flow cooling → Heat-up air 8×
Heat-up air → Heat-up N_2+H_2 11×
Shutdown → Heat-up N_2+H_2 2×
Heat-up N_2+H_2 → Hot standby 9×
Shutdown → Hot standby 7×
Hot standby → Fuel switch 18×
Fuel switch → Part load 10×

Shutdown → Heat-up air

3 times

few cases: read as examples

A cold start begins. The air comes on (0 → 6000 ln/min); 5–8 min later the gas-line heaters (≈ 4 A, recycle heater ≈ 8 A) and the air heater follow. The stack first dips a little from the cold air, then starts warming.



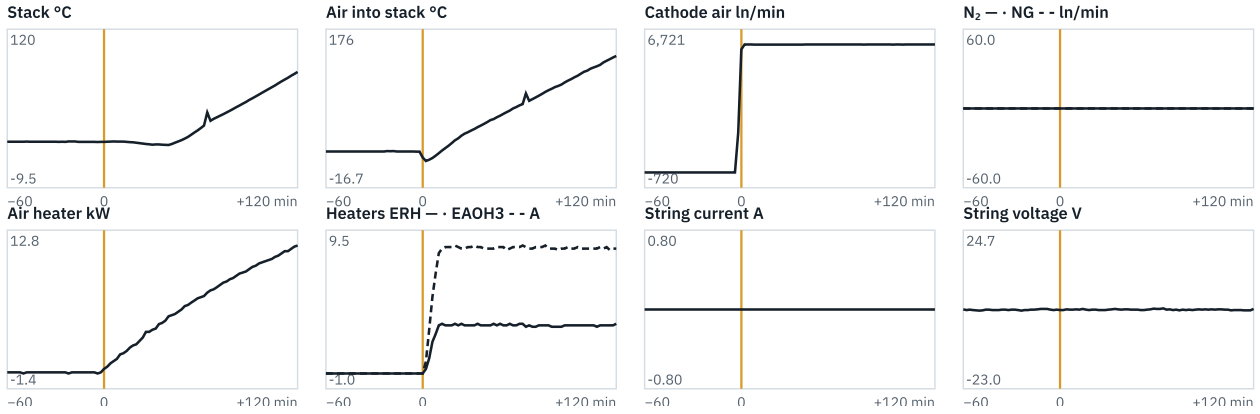
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Stack temperature	154	126	196	within 1 min
▼ Air into stack (TIRC 2.06)	153	104	278	within 1 min
▲ Cathode air, both streams	0.0	5,999	6,000	within 1 min
▲ Air heater EAPH power	0.0	2.6	18.0	8 min
▲ Gas-line heater ERH current	0.0	4.4	4.2	5 min
▲ Recycle heater EAOH 3 current	0.0	8.3	8.2	5 min
▲ After-burner (TIRC 5.09)	122	126	181	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	12.5	26.7	31.2	within 1 min

No real change: Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage, Reformer catalyst inlet.

Air-flow cooling → **Heat-up air** 8 times

A restart while the cooling air was still blowing and the stack was already cold (≈ 28 °C). The air steps to 6000 ln/min and the air heater and gas-line heaters switch on within 1–2 min. The air into the stack rises first; the stack follows a few minutes later.



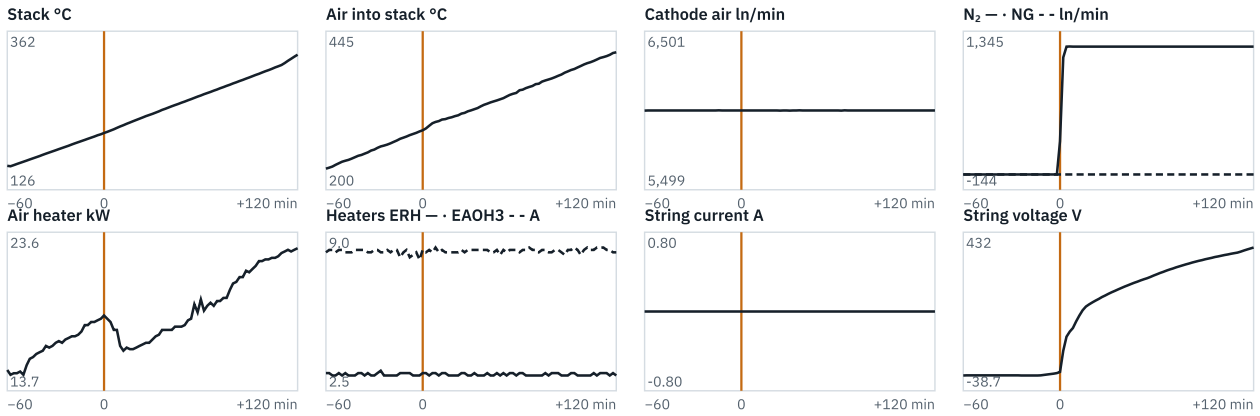
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Stack temperature	28.0	25.9	212	3 min
▲ Air into stack (TIRC 2.06)	25.9	52.6	284	within 1 min
▲ Cathode air, both streams	2,490	5,999	6,000	within 1 min
▲ Air heater EAPH power	0.0	3.9	15.0	within 1 min
▲ Gas-line heater ERH current	0.0	3.2	3.1	2 min
▲ Recycle heater EAOH 3 current	0.0	8.4	8.2	2 min
▲ After-burner (TIRC 5.09)	15.0	25.5	198	within 1 min
▲ Reformer catalyst inlet (TIRC 3.07)	13.5	22.0	174	4 min
▲ Exhaust safety (TIRZ 8.1.10)	13.5	18.8	31.9	within 1 min

No real change: Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage.

Heat-up air → **Heat-up N_2+H_2** 11 times

At a stack of ≈ 204 °C nitrogen (1200 ln/min) and hydrogen (60 ln/min) switch on in the same minute. The stack voltage appears: ≈ 10 V before, ≈ 250 V after 30 min, ≈ 430 V by the end of the stage. Air and heaters do not change. Cold forming gas cools the reformer inlet (169 → 54 °C).



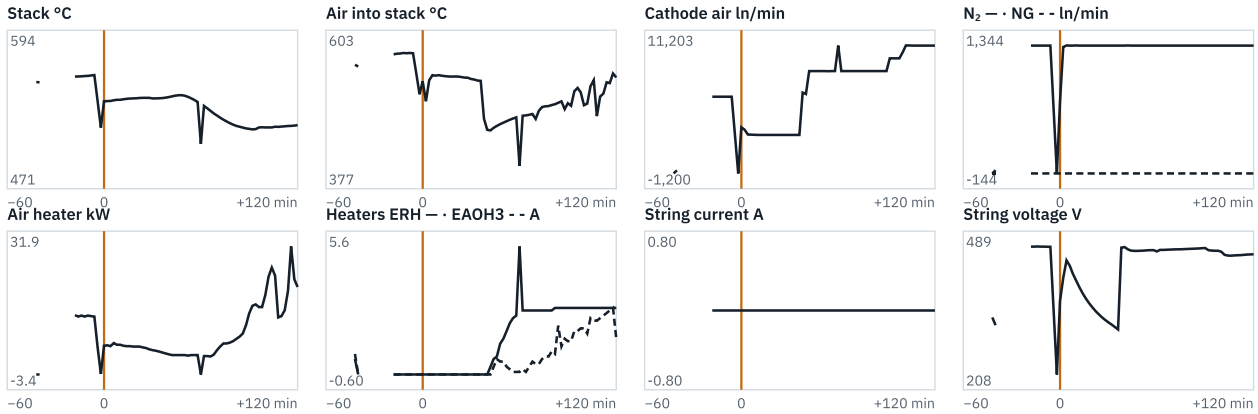
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Stack temperature	204	239	495	within 1 min
▲ Air into stack (TIRC 2.06)	284	322	500	within 1 min
▲ Nitrogen N_2 (FIRC 1.3)	0.0	1,200	1,200	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	60.0	60.0	within 1 min
▲ String 1 stack voltage	9.6	249	432	within 1 min
▲ Air heater EAPH power	17.9	18.1	13.1	within 1 min
▲ After-burner (TIRC 5.09)	185	263	458	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	169	54.3	93.2	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	32.2	50.1	101	3 min

No real change: Cathode air, both streams, Natural gas, Reformer air, String 1 current, Gas-line heater ERH current, Recycle heater EAOH 3 current.

Shutdown → Heat-up N_2+H_2 2 times few cases: read as examples

A restart of a stack that was stopped while still hot (≈ 518 °C). Air, N_2 and H_2 come back together and the air heater climbs back to ≈ 13 kW.



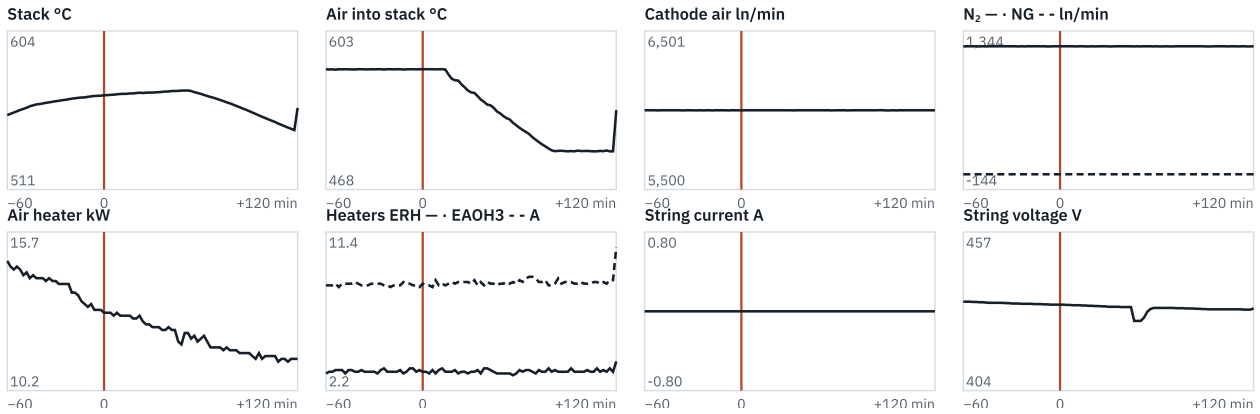
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Stack temperature	518	541	570	within 1 min
▲ Air into stack (TIRC 2.06)	512	534	560	within 1 min
▲ Cathode air, both streams	0.0	3,004	8,005	within 1 min
▲ Nitrogen N_2 (FIRC 1.3)	0.0	1,200	1,200	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	60.0	60.0	within 1 min
▲ String 1 stack voltage	235	327	448	within 1 min
▲ Air heater EAPH power	0.10	5.4	13.0	within 1 min
▼ Gas-line heater ERH current	0.0	0.0	2.6	within 1 min
▼ Recycle heater EAOH 3 current	0.0	0.0	5.4	within 1 min
▲ After-burner (TIRC 5.09)	390	417	535	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	384	114	134	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	39.5	185	118	within 1 min

No real change: Natural gas, Reformer air, String 1 current.

● **Heat-up N_2+H_2** → ● **Hot standby** 9 times

A quiet border: nothing is switched. The stack simply stops climbing (566 → 568 °C after 30 min), and the air heater eases from 13 to 9 kW as it only has to hold the temperature.



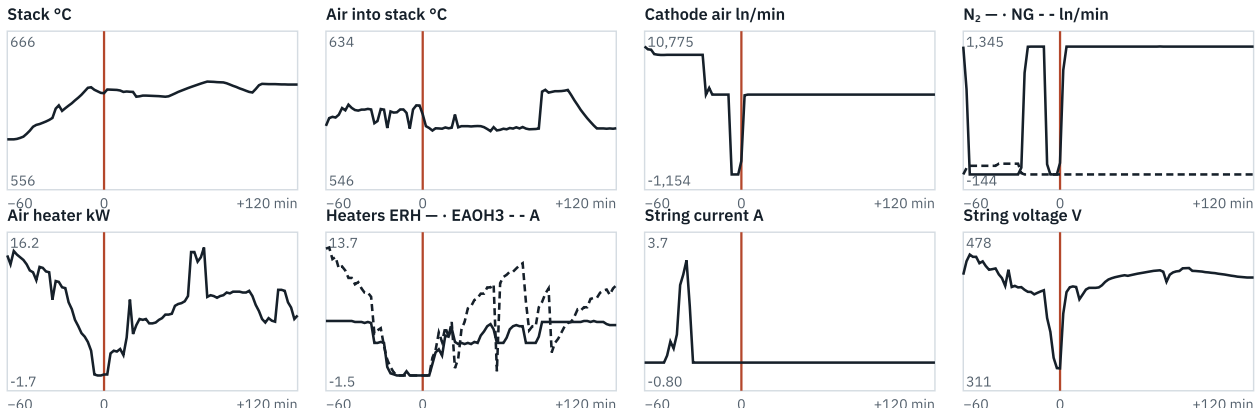
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Air heater EAPH power	13.1	12.4	9.3	within 1 min

No real change: Stack temperature, Air into stack, Cathode air, both streams, Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage, Gas-line heater ERH current, Recycle heater EAOH 3 current, After-burner, Reformer catalyst inlet, Exhaust safety.

Shutdown → Hot standby 7 times

Recovery after a hot stop (stack ≈ 628 °C, typically after an emergency stop). Air (6000) and N_2 + H_2 return within 1 min, the air heater and gas-line heaters within 2–4 min. The voltage recovers from 339 to 425 V.



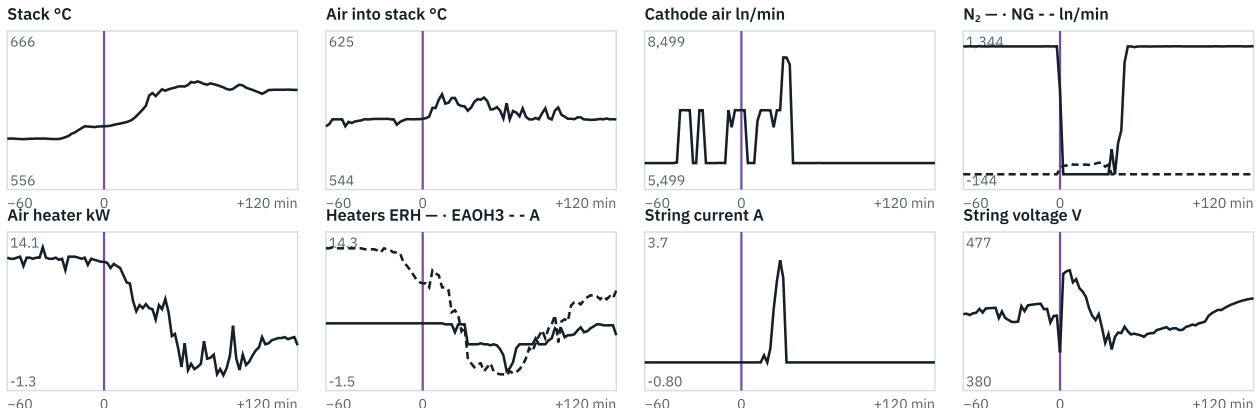
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Cathode air, both streams	0.0	6,000	9,722	within 1 min
▲ Nitrogen N_2 (FIRC 1.3)	0.0	1,200	1,200	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	60.0	60.0	within 1 min
▲ String 1 stack voltage	339	425	426	within 1 min
▲ Air heater EAPH power	0.0	6.9	12.8	2 min
▲ Gas-line heater ERH current	0.0	3.1	5.2	2 min
▲ Recycle heater EAOH 3 current	0.0	6.7	8.7	4 min
▼ After-burner (TIRC 5.09)	619	570	563	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	566	182	153	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	96.6	143	122	within 1 min

No real change: Stack temperature, Air into stack, Natural gas, Reformer air, String 1 current.

Hot standby → **Fuel switch** 18 times

N_2 and H_2 go off and natural gas (≈ 98 ln/min) with reformer air (≈ 360 ln/min) come on in the same minute. Within 1 min the reformer catalyst jumps from 157 to 752 °C and the after-burner from 561 to 708 °C. The voltage rises from 426 to 465 V. The stack stays near 600 °C.



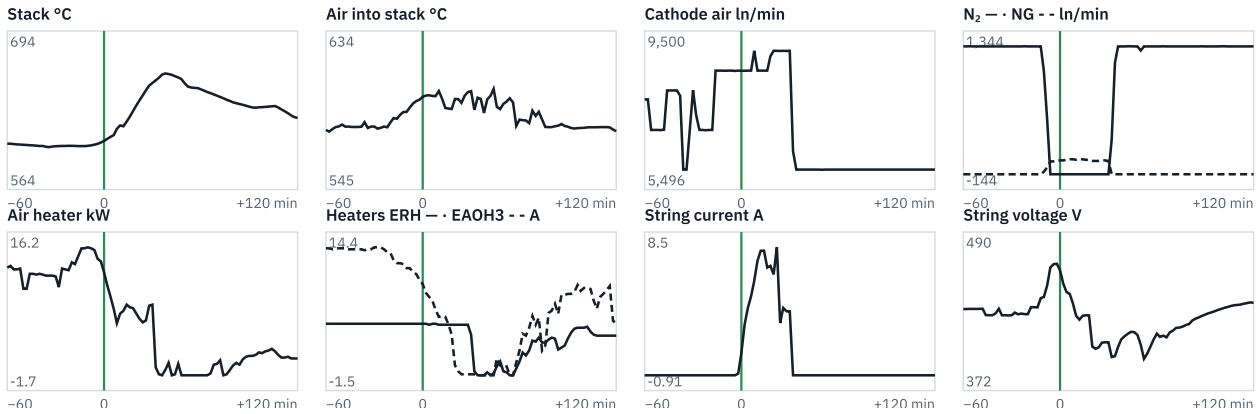
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Nitrogen N_2 (FIRC 1.3)	1,200	0.0	0.0	within 1 min
▼ Hydrogen H_2 (FIRC 1.5)	60.0	0.0	0.0	within 1 min
▲ Natural gas (FIRC 1.4)	0.0	97.9	97.9	within 1 min
▲ Reformer air (FIRC 1.2)	0.0	358	386	within 1 min
▲ String 1 stack voltage	426	465	464	within 1 min
▲ Recycle heater EAOH 3 current	9.2	9.8	10.6	within 1 min
▲ After-burner (TIRC 5.09)	561	708	682	within 1 min
▲ Reformer catalyst inlet (TIRC 3.07)	157	752	767	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	130	116	116	within 1 min

No real change: Stack temperature, Air into stack, Cathode air, both streams, String 1 current, Air heater EAPH power, Gas-line heater ERH current.

Fuel switch → **Part load** 10 times

The current is switched on (0 → ≈ 9 A per string): this is the border. Under load the voltage drops from 466 to 431 V. Natural gas goes up from 120 to 140 ln/min. The recycle heater (10.6 → 2.3 A) and the air heater (14 → 7 kW) back off, because the stack now makes its own heat; the stack rises from 600 to 644 °C in 30 min.



Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Stack temperature	600	644	636	within 1 min
▲ Cathode air, both streams	8,500	9,000	8,500	within 1 min
▲ Natural gas (FIRC 1.4)	120	140	140	within 1 min
▼ Reformer air (FIRC 1.2)	440	310	308	within 1 min
▲ String 1 current	0.0	8.7	8.4	within 1 min
▼ String 1 stack voltage	466	431	431	within 1 min
▼ Air heater EAPH power	14.1	6.8	5.5	within 1 min
▼ Recycle heater EAOH 3 current	10.6	2.3	4.7	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	785	713	711	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	115	155	139	within 1 min

No real change: Air into stack, Nitrogen N_2, Hydrogen H_2, Gas-line heater ERH current, After-burner.

If you remember one thing: Supplies switch in the same minute; temperatures take minutes to hours to follow.

Unloading, cooling and trips

On the way down each move **switches supplies off**, often several in the same minute. Trips switch everything except air, N_2 and H_2 off at once; cooling then takes many hours because the hot box holds its heat.

How to read each change

Before = median of the last 15 min of the old stage · Early in next stage = about 30 min in (or the end of the next stage, if it is shorter) · End of next stage = its last 15 min · Starts to move = minutes until the reading has made a tenth of its move. ▲ rises, ▼ falls. Only readings that really change are listed.

Fuel switch → Hot standby 7×

Part load → Hot standby 3×

Hot standby → Air-flow cooling 6×

Air-flow cooling → Shutdown 12×

Shutdown → Air-flow cooling 14×

Heat-up air → Shutdown 2×

Heat-up N_2+H_2 → Shutdown 3×

Part load → Air-flow cooling 3×

Part load → Trip / Emergency purge 2×

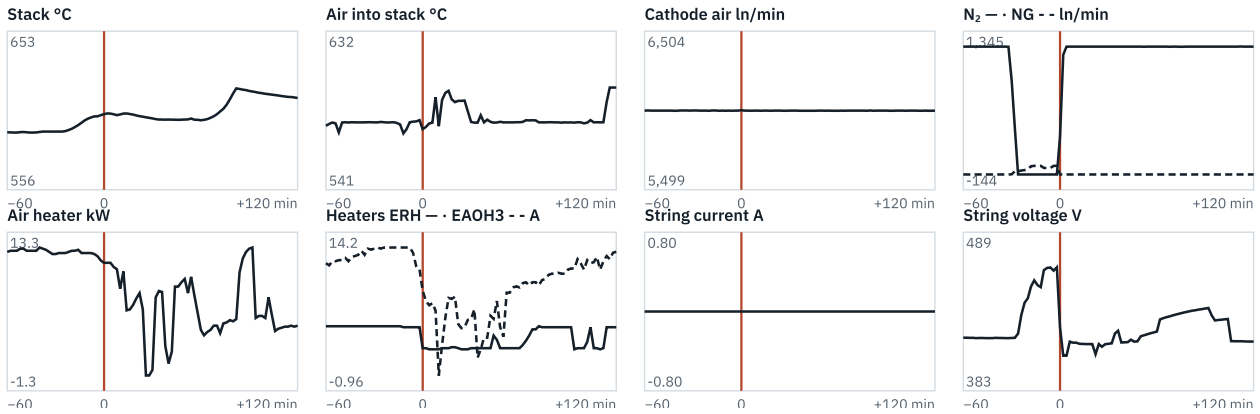
Air-flow cooling → Trip / Emergency purge 2×

Trip / Emergency purge → Air-flow cooling 2×

Trip / Emergency purge → Shutdown 3×

Fuel switch → Hot standby 7 times

The gas start is called off. Natural gas and reformer air go off and N_2 + H_2 come back on in the same minute. The reformer cools from 730 to 192 °C and the after-burner from 643 to 584 °C. The voltage falls from 465 to 415 V and the heaters step back down (5.1 → 3.3 A, recycle 12.7 → 8.1 A).



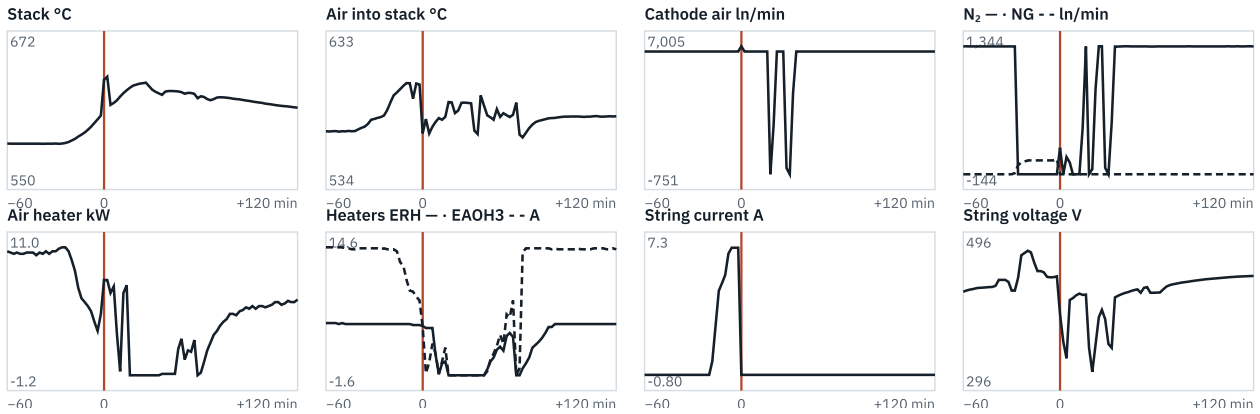
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Nitrogen N_2 (FIRC 1.3)	0.0	1,200	1,200	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	60.0	60.0	within 1 min
▼ Natural gas (FIRC 1.4)	80.0	0.0	0.0	within 1 min
▼ Reformer air (FIRC 1.2)	304	0.0	0.0	within 1 min
▼ String 1 stack voltage	465	415	421	within 1 min
▼ Air heater EAPH power	11.3	9.4	7.2	within 1 min
▼ Gas-line heater ERH current	5.1	3.3	3.4	within 1 min
▼ Recycle heater EAOH 3 current	12.7	8.1	8.0	within 1 min
▼ After-burner (TIRC 5.09)	643	584	588	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	730	192	206	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	120	140	139	within 1 min

No real change: Stack temperature, Air into stack, Cathode air, both streams, String 1 current.

● **Part load** → ● **Hot standby** 3 times `few cases: read as examples`

Current and gas go off together (5.6 → 0 A). The voltage drops from 441 to 345 V and the reformer cools from 809 to 428 °C, but the stack stays hot (≈ 637 °C) on stored heat. In these 3 cases the forming gas had not come back yet in the short hot standby that followed.



Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

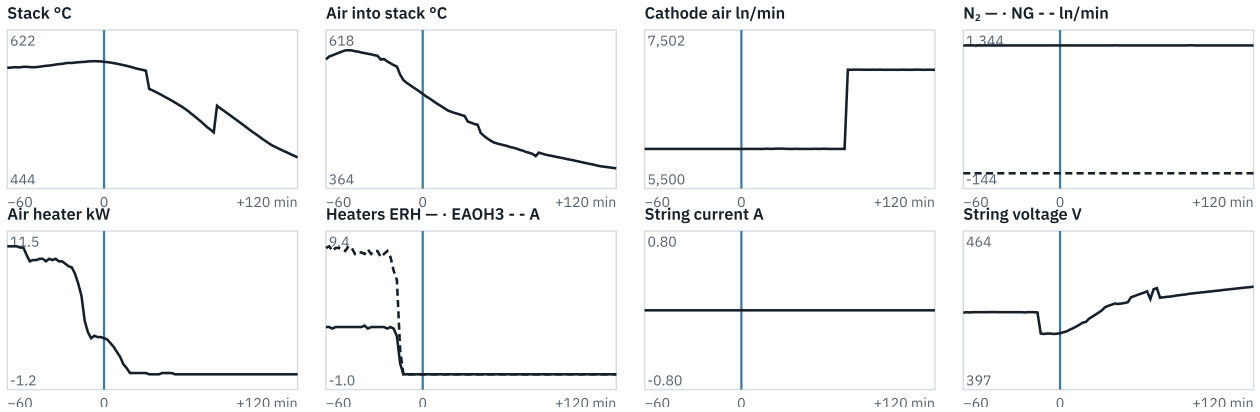
READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Stack temperature	599	637	637	within 1 min
▼ Natural gas (FIRC 1.4)	130	0.0	0.0	within 1 min
▼ Reformer air (FIRC 1.2)	507	0.0	0.0	within 1 min
▼ String 1 current	5.6	0.0	0.0	within 1 min
▼ String 1 stack voltage	441	345	351	within 1 min
▲ Air heater EAPH power	3.6	6.6	6.8	within 1 min
▼ Gas-line heater ERH current	5.2	4.8	4.8	within 1 min
▼ Recycle heater EAOH 3 current	8.4	2.3	1.6	within 1 min
▼ After-burner (TIRC 5.09)	756	623	629	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	809	428	435	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	121	178	172	within 1 min

No real change: Air into stack, Cathode air, both streams, Nitrogen N_2, Hydrogen H_2.

Hot standby → **Air-flow cooling** 6 times

The cool-down starts. The air heater (≈ 3 kW → 0) and the gas-line heaters switch off; the air keeps blowing at 6000 ln/min and is often raised to 7000–10000 later. N_2 + H_2 stay on

while the stack is above 210 °C. The stack falls from 587 °C to 575 °C after 30 min and to ≈ 268 °C by the end (median 7.5 h).



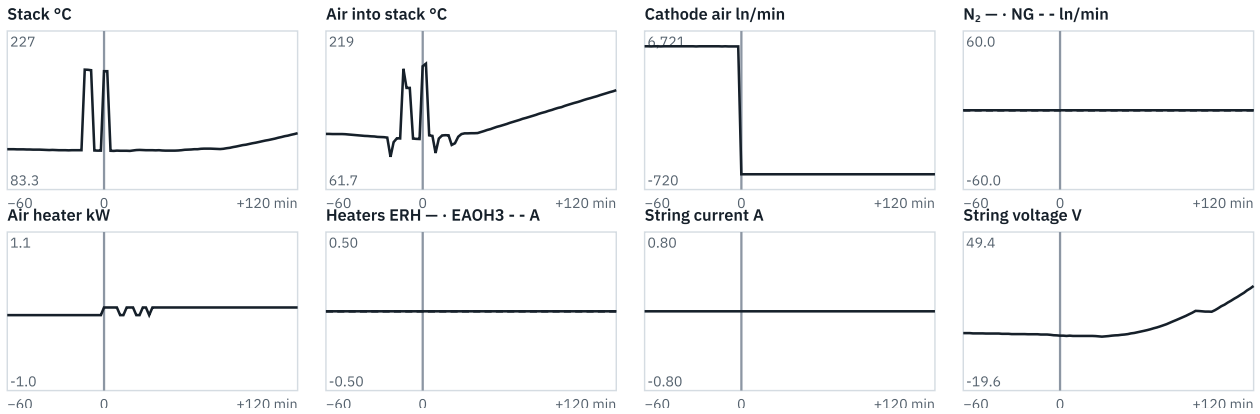
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Stack temperature	587	575	268	3 min
▼ Air into stack (TIRC 2.06)	529	477	246	within 1 min
▲ Cathode air, both streams	6,000	6,003	7,001	within 1 min
▲ String 1 stack voltage	421	433	309	2 min
▼ Air heater EAPH power	2.9	0.10	0.0	within 1 min
▼ After-burner (TIRC 5.09)	542	527	275	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	140	134	64.7	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	117	112	85.4	within 1 min

No real change: Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, Gas-line heater ERH current, Recycle heater EAOH 3 current.

Air-flow cooling → **Shutdown** 12 times

The air is switched off (6000 → 0) once the stack is down to ≈ 118 °C. Nothing else changes, because everything else is already off. From then on the stack drifts down very slowly (118 → 111 °C).



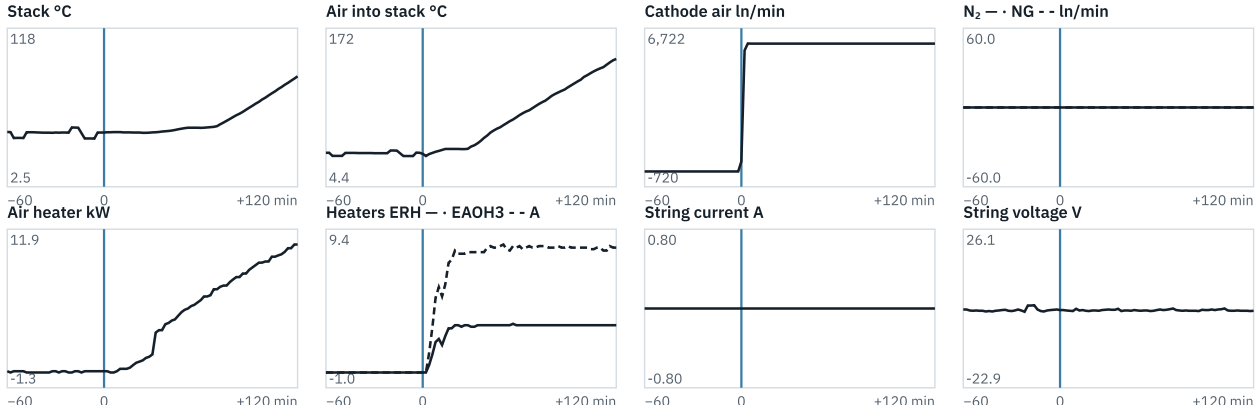
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Cathode air, both streams	6,000	0.0	0.0	within 1 min
▲ Reformer catalyst inlet (TIRC 3.07)	46.4	63.8	90.4	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	32.8	19.6	18.6	within 1 min

No real change: Stack temperature, Air into stack, Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage, Air heater EAPH power, Gas-line heater ERH current, Recycle heater EAOH 3 current, After-burner.

Shutdown → Air-flow cooling 14 times

Air comes back on at a low rate (≈ 2500 ln/min) in a cold plant (≈ 42 °C), usually for only a few minutes (median 2.5 min): ventilation or a check before a restart.



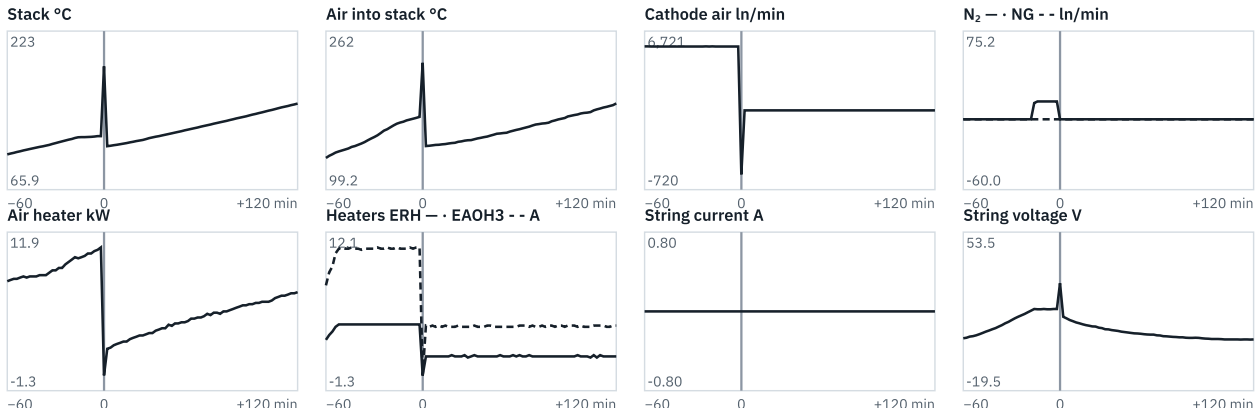
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Cathode air, both streams	0.0	2,490	2,490	within 1 min

No real change: Stack temperature, Air into stack, Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage, Air heater EAPH power, Gas-line heater ERH current, Recycle heater EAOH 3 current, After-burner, Reformer catalyst inlet, Exhaust safety.

Heat-up air → **Shutdown** 2 times few cases: read as examples

An aborted start-up during the air-only heat-up: air, air heater and gas-line heaters all go off at once.



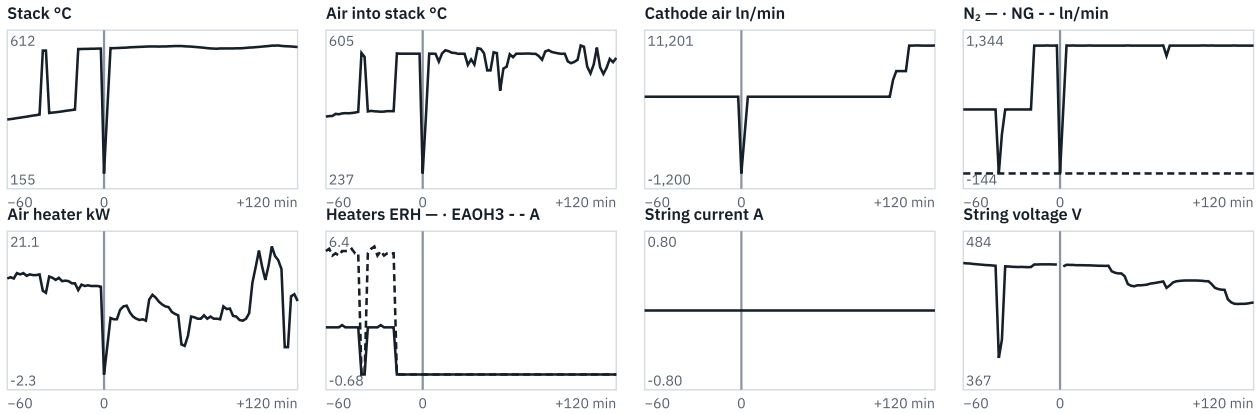
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Stack temperature	118	190	170	within 1 min
▲ Air into stack (TIRC 2.06)	171	211	169	within 1 min
▼ Cathode air, both streams	6,000	0.0	0.0	within 1 min
▼ Air heater EAPH power	10.1	0.0	0.0	within 1 min
▼ Gas-line heater ERH current	4.3	0.0	0.0	within 1 min
▼ Recycle heater EAOH 3 current	10.7	0.0	0.0	within 1 min
▲ After-burner (TIRC 5.09)	112	196	157	within 1 min
▲ Reformer catalyst inlet (TIRC 3.07)	103	200	160	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	25.8	18.0	7.8	within 1 min

No real change: Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current, String 1 stack voltage.

Heat-up N_2+H_2 → **Shutdown** 3 times `few cases: read as examples`

An aborted start-up with forming gas on: air, N_2, H_2 and the air heater all go off at once. The stack (≈ 559 °C) then cools slowly without air.



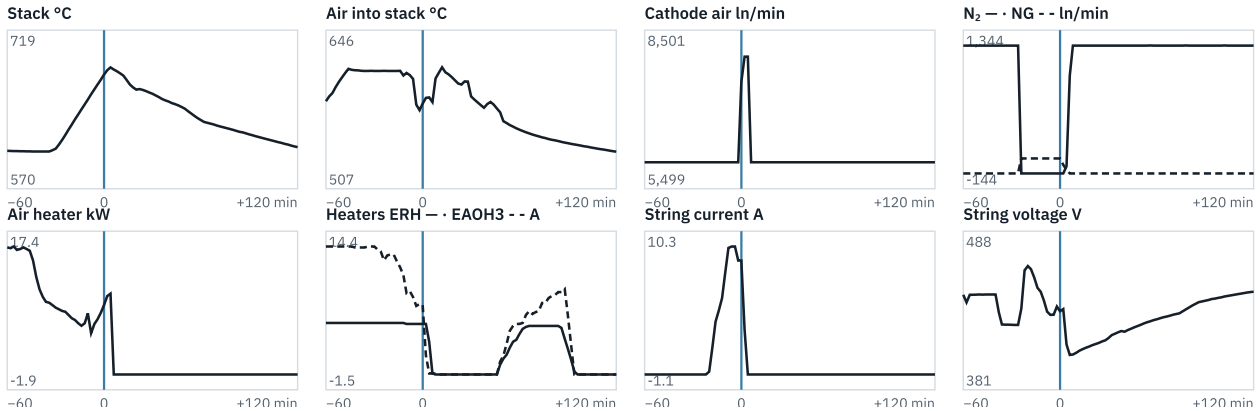
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Stack temperature	559	197	181	within 1 min
▼ Air into stack (TIRC 2.06)	550	227	180	within 1 min
▼ Cathode air, both streams	6,000	0.0	0.0	within 1 min
▼ Nitrogen N_2 (FIRC 1.3)	1,200	0.0	0.0	within 1 min
▼ Hydrogen H_2 (FIRC 1.5)	60.0	0.0	0.0	within 1 min
String 1 stack voltage	460	–	–	–
▼ Air heater EAPH power	13.0	0.0	0.0	within 1 min
▼ After-burner (TIRC 5.09)	519	179	144	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	121	24.1	14.6	within 1 min

No real change: Natural gas, Reformer air, String 1 current, Gas-line heater ERH current, Recycle heater EAOH 3 current, Reformer catalyst inlet.

Part load → **Air-flow cooling** 3 times `few cases: read as examples`

Few and mixed cases. In these 3 short stretches the air changed while gas and current were still flowing: hand-overs during unloading rather than real cool-downs.



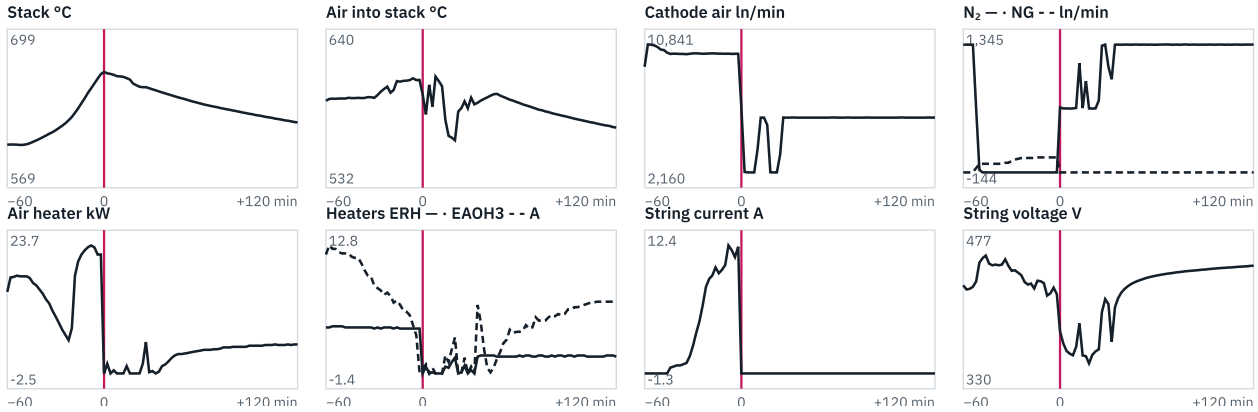
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Cathode air, both streams	6,000	6,765	6,765	within 1 min
▼ Reformer air (FIRC 1.2)	383	308	308	within 1 min
▼ String 1 current	9.1	8.0	8.0	within 1 min
▲ Air heater EAPH power	5.0	6.0	6.0	within 1 min
▼ Recycle heater EAOH 3 current	8.4	6.3	6.3	within 1 min
▲ Exhaust safety (TIRZ 8.1.10)	146	194	193	within 1 min

No real change: Stack temperature, Air into stack, Nitrogen N_2, Hydrogen H_2, Natural gas, String 1 stack voltage, Gas-line heater ERH current, After-burner, Reformer catalyst inlet.

Part load → **Trip / Emergency purge** 2 times `few cases: read as examples`

A safety cut while making power. Current, natural gas, reformer air, the air heater and the gas-line heaters all go off at once, and the N_2 + H_2 purge starts. The after-burner drops from 751 to 633 °C.



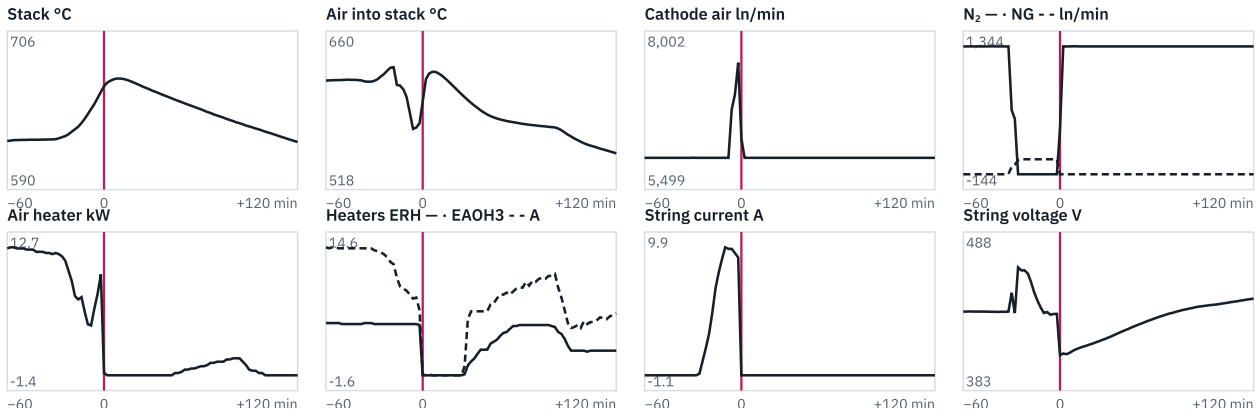
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Cathode air, both streams	9,496	4,706	4,706	within 1 min
▲ Nitrogen N_2 (FIRC 1.3)	0.0	792	792	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	39.6	39.6	within 1 min
▼ Natural gas (FIRC 1.4)	140	0.0	0.0	within 1 min
▼ Reformer air (FIRC 1.2)	244	0.0	0.0	within 1 min
▼ String 1 current	10.2	0.0	0.0	within 1 min
▼ String 1 stack voltage	422	397	396	within 1 min
▼ Air heater EAPH power	20.6	0.0	0.0	within 1 min
▼ Gas-line heater ERH current	4.0	0.20	0.0	within 1 min
▼ Recycle heater EAOH 3 current	5.2	0.30	0.0	within 1 min
▼ After-burner (TIRC 5.09)	751	633	640	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	692	360	368	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	188	170	171	within 1 min

No real change: Stack temperature, Air into stack.

Air-flow cooling → **Trip / Emergency purge** 2 times `few cases: read as examples`

A safety cut in minutes labelled air-flow cooling but with gas (140 ln/min) and current (≈ 8 A) still on, i.e. during unloading. Same pattern as a trip from part load: everything off except air, N_2 and H_2.



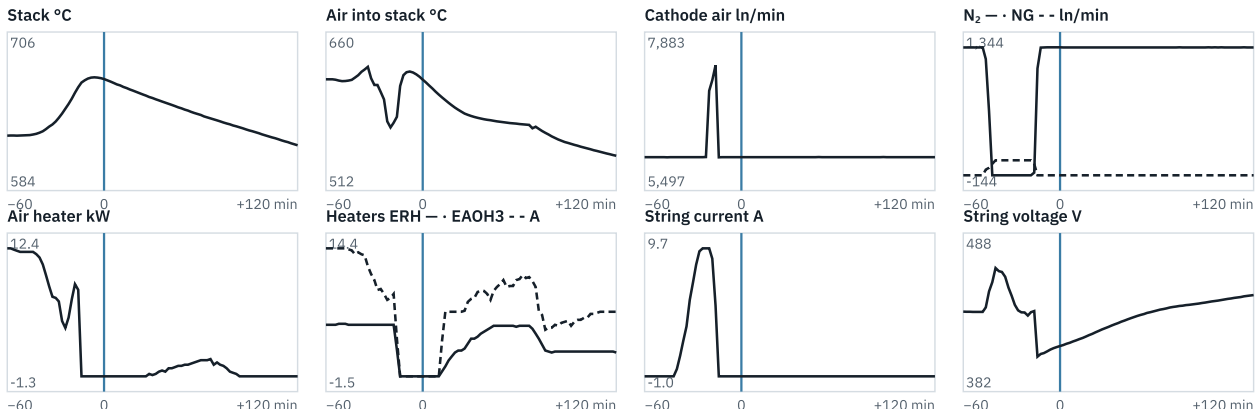
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▲ Air into stack (TIRC 2.06)	576	619	621	within 1 min
▼ Cathode air, both streams	7,383	6,000	6,000	within 1 min
▲ Nitrogen N_2 (FIRC 1.3)	0.0	1,200	1,200	within 1 min
▲ Hydrogen H_2 (FIRC 1.5)	0.0	60.0	60.0	within 1 min
▼ Natural gas (FIRC 1.4)	140	0.0	0.0	within 1 min
▼ Reformer air (FIRC 1.2)	349	0.0	0.0	within 1 min
▼ String 1 current	8.2	0.0	0.0	within 1 min
▼ String 1 stack voltage	434	412	409	within 1 min
▼ Air heater EAPH power	7.9	0.0	0.0	within 1 min
▼ Gas-line heater ERH current	5.1	0.0	0.0	within 1 min
▼ Recycle heater EAOH 3 current	7.5	0.0	0.0	within 1 min
▼ After-burner (TIRC 5.09)	763	620	634	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	752	223	240	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	195	163	166	within 1 min

No real change: Stack temperature.

Trip / Emergency purge → **Air-flow cooling** 2 times few cases: read as examples

After the 15-min purge the stack is cooled by air (671 → 655 °C after 30 min → 453 °C at the end). The gas-line heaters come back on 19–27 min after the border.



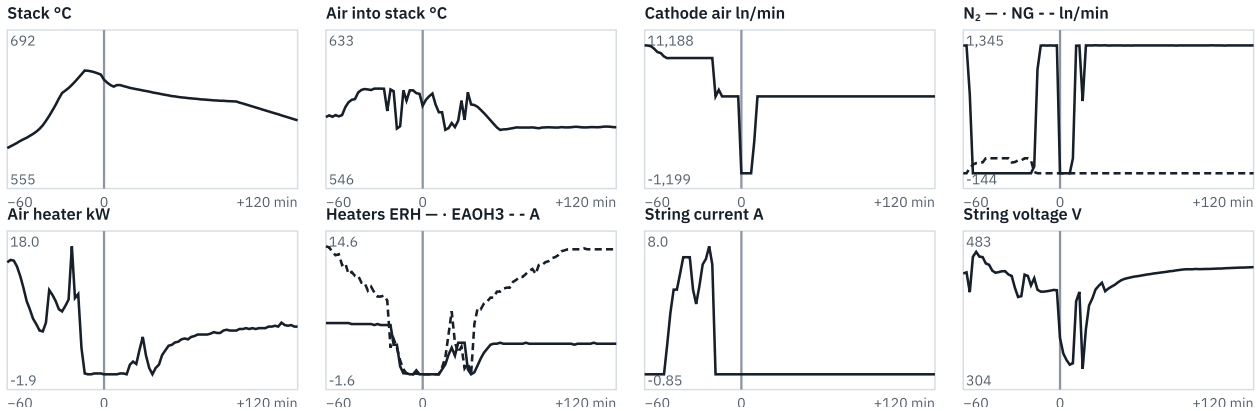
Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Stack temperature	671	655	453	2 min
▼ Air into stack (TIRC 2.06)	621	582	419	within 1 min
▲ String 1 stack voltage	409	425	491	within 1 min
▲ Air heater EAPH power	0.0	0.20	3.0	27 min
▲ Gas-line heater ERH current	0.0	3.6	2.5	19 min
▲ Recycle heater EAOH 3 current	0.0	7.8	6.4	19 min
▼ After-burner (TIRC 5.09)	634	591	436	within 1 min
▼ Reformer catalyst inlet (TIRC 3.07)	240	176	93.5	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	166	149	103	within 1 min

No real change: Cathode air, both streams, Nitrogen N_2, Hydrogen H_2, Natural gas, Reformer air, String 1 current.

Trip / Emergency purge → **Shutdown** 3 times few cases: read as examples

After the purge all flows stop (air, N_2 and H_2 → 0). The stack stays hot (≈ 646 °C) with no air.



Median of all cases, from 60 min before to 120 min after the move (coloured line = the move). The next stage may already have ended inside this window.

READING	BEFORE	EARLY IN NEXT STAGE	END OF NEXT STAGE	STARTS TO MOVE
▼ Cathode air, both streams	6,000	0.0	0.0	within 1 min
▼ Nitrogen N_2 (FIRC 1.3)	1,200	0.0	0.0	within 1 min
▼ Hydrogen H_2 (FIRC 1.5)	60.0	0.0	0.0	within 1 min
▼ String 1 stack voltage	414	338	343	within 1 min
▼ After-burner (TIRC 5.09)	622	524	525	within 1 min
▲ Reformer catalyst inlet (TIRC 3.07)	243	432	421	within 1 min
▼ Exhaust safety (TIRZ 8.1.10)	184	114	119	within 1 min

No real change: Stack temperature, Air into stack, Natural gas, Reformer air, String 1 current, Air heater EAPH power, Gas-line heater ERH current, Recycle heater EAOH 3 current.

If you remember one thing: Going down, things switch off at once; the hot stack keeps its heat for hours.